

Eleven Primitives and Three Gates: The Universal Structure of Computational Imaging

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Abstract

Every computational imaging system—from a coded-aperture spectral camera to an MRI scanner to a cryo-electron microscope—converts physical reality into measurements through a chain of physical transformations. Despite the apparent diversity of these systems across carrier families, length scales, and modalities, here we show that the space of all such forward models has *finite structural complexity*. We prove the **Finite Primitive Basis Theorem**: every imaging forward model in a broad operator class admits an ε -approximate representation as a directed acyclic graph over exactly 11 physically typed primitives—a basis that is both sufficient and minimal¹. This is a discovery, not a design choice: the primitive library saturates at 11 as modality coverage grows past 168 registered systems across 19 categories, with no new primitive required for the most recent 138 additions, proving that the structural complexity of imaging physics is bounded. We further establish the **Triad Decomposition**, a universal diagnostic law governing every failure in computational image recovery: reconstruction error decomposes into exactly three independent root causes—information deficiency, carrier noise, and operator mismatch—mirroring the role of entropy, thermal noise, and free-energy barriers in the thermodynamics of irreversible processes. Hardware validation across twelve modalities spanning all five carrier families—photons, electrons, X-rays, nuclear spins, and acoustic waves—including cryo-electron microscopy and clinical cone-beam CT, confirms that operator mismatch, not information deficiency or noise, is the dominant reconstruction bottleneck under standard operating conditions, recoverable by +0.8 to +10.7 dB (± 0.3 dB, 95% bootstrap confidence interval) through forward-model correction alone—often exceeding the gain from upgrading a classical solver to a state-of-the-art deep network. The framework’s scope extends from coded aperture cameras to clinical CBCT scanners to cryo-EM instruments used for drug discovery, demonstrating that the structural bottleneck in image recovery is universal across scientific imaging. These results shift the primary locus of investment in computational imaging from solver design to calibration, and provide a unifying blueprint for decomposing forward models across physical measurement science.

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Introduction

Every imaging instrument—whether it captures spectra, spins, X-rays, or sound waves—converts a physical scene into digital measurements through a chain of transformations: a wave propagates, interacts with the object, passes through optics, and is recorded by a detector. Computational imaging reconstructs the scene by inverting this chain. The reconstruction algorithm assumes a mathematical model of the chain (the “forward model”), but this model inevitably diverges from reality: optics shift, detectors drift, and calibration degrades. When the model is wrong, the reconstruction fails—not because the algorithm is weak, but because it is solving the wrong problem.

This paper establishes that the space of imaging forward models has a surprisingly simple and universal structure, and that exploiting this structure reveals operator mismatch as the systematic, cross-modality bottleneck in reconstruction quality. Over the past decade, the community has invested enormous effort in improving reconstruction algorithms—progressing from compressed sensing^{2,3} and plug-and-play priors^{4,5} to deep unrolling networks⁶ and vision transformers⁷—with snapshot compressive imaging emerging as a unifying framework for coded aperture systems⁸—yet these algorithms routinely fail when deployed on real instruments⁹. Calibration methods exist for specific instruments—ESPIRiT¹⁰ for MRI, ePIE¹¹ for ptychography—but they do not generalise across modalities, and no systematic framework decomposes reconstruction failures into root causes.

Here we establish two complementary foundations within Physics World Models (PWM). The **Finite Primitive Basis Theorem** (Theorem 3) proves that every imaging forward model in a broad operator class $\mathcal{C}_{\text{img}}$ admits an ε -approximate representation as a typed directed acyclic graph (DAG) over exactly 11 canonical primitives—and that this library is minimal¹. The **Triad Decomposition** proves that every reconstruction failure decomposes into three gates—information deficiency (**Gate 1**), carrier noise (**Gate 2**), and operator mismatch (**Gate 3**)—with a formal condition (Theorem 5) under which **Gate 3** dominates. The 11 primitives *enable* the 3-gate diagnosis: the DAG structure localizes mismatch to a specific physical operator, making correction actionable.

We validate both results across twelve modalities spanning all five carrier families—optical photons (CASSI¹², CACTI¹³, SPC¹⁴, lensless, compressive holography, fluorescence microscopy), X-ray photons (CT¹⁵, cone-beam CT), electrons (ptychography¹⁶, cryo-EM), nuclear spins (MRI^{17,18}), and acoustic waves (ultrasound)—with hardware validation on ten modalities confirming Triad predictions on physical measurements. The validated modalities include two Nobel Prize-winning techniques (cryo-EM, 2017 Chemistry; super-resolution fluorescence, 2014 Chemistry), three clinically deployed instruments (CT, CBCT, MRI), and the first acoustic-carrier validation in computational imaging. A held-out closure test on 8 additional modalities confirms basis completeness with saturating growth. The proof of Theorem 3 is in Supplementary Note 12; formal primitive semantics and typed DAG denotation are developed in¹.

72 **Why this matters beyond computational imaging.** Any physical measurement system—
 73 seismographs, telescopes, particle detectors, environmental sensor networks—faces the same
 74 structural problem: a forward model maps the quantity of interest to observations, and
 75 model fidelity limits what can be recovered. The finding that eleven primitives span the
 76 entire space of imaging forward models raises a fundamental question about the grammar
 77 of physical measurement: if five carrier families, twelve validated modalities, and 168 regis-
 78 tered systems across 19 categories all compose from the same small set of operators, is this
 79 a contingent feature of imaging, or does it reflect a deeper constraint on how information
 80 flows from physical systems to digital observations? The OPERATORGRAPH formalism pro-
 81 vides a template for decomposing forward models in other domains into typed, composable
 82 primitives with validated adjoints. The Triad Decomposition—separating information ca-
 83 pacity, noise, and model fidelity as independent failure modes—applies wherever an inverse
 84 problem is solved computationally. The practical lesson is domain-general: before investing
 85 in a better solver, diagnose whether the model is the bottleneck.

86 The Finite Primitive Basis

87 A foundational question in computational imaging is whether the diversity of imaging for-
 88 ward models—from coded aperture cameras to MRI scanners to electron microscopes—can
 89 be captured by a small, fixed set of primitive operators. We answer affirmatively.

90 **The OperatorGraph representation.** Every imaging forward model is encoded as a
 91 typed directed acyclic graph (DAG) in which each node wraps a single primitive physical
 92 operator and edges define the data flow from source to detector. Every primitive implements
 93 both a `forward()` and an `adjoint()` method, with validated adjoint consistency ensuring
 94 $\langle H\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle$ to within numerical precision. Each edge carries tensor shape and
 95 dtype metadata, enabling static validation before execution. We call this formalism the
 96 OPERATORGRAPH intermediate representation (IR).

97 **Eleven canonical primitives.** The OPERATORGRAPH IR defines a library of exactly 11
 98 canonical primitives $\mathcal{B} = \{P, M, \Pi, F, C, \Sigma, D, S, W, R, \Lambda\}$:
 99 The Detect nonlinearity η is restricted to five canonical families (linear-field: $\eta(x) = gx$;
 100 logarithmic; sigmoid; intensity-square-law: $\eta(x) = g|x|^2$; coherent-field), each with at most
 101 2 scalar parameters. The Transform primitive Λ applies a pointwise nonlinear function
 102 within the physics chain (not at the detector) and is restricted to five families: exponential
 103 attenuation ($e^{-\alpha x}$, Beer–Lambert), logarithmic compression, phase wrapping ($\arg(e^{ix})$),
 104 polynomial ($\sum a_k x^k$, $d \leq 5$), and saturation. Both Detect and Transform are pointwise
 105 and parameter-limited, preventing either from becoming a universal approximator (see¹ for
 106 formal definitions and non-universality proofs).

#	Primitive	Notation	Physical action
1	Propagate	$P(d, \lambda)$	Free-space wave propagation
2	Modulate	$M(\mathbf{m})$	Element-wise multiplication (mask, coil, absorption)
3	Project	$\Pi(\theta)$	Radon line-integral projection
4	Encode	$F(\mathbf{k})$	Fourier-domain encoding (k -space)
5	Convolve	$C(\mathbf{h})$	Spatial convolution (PSF)
6	Accumulate	Σ	Summation over spectral/temporal axis
7	Detect	$D(g, \eta)$	Detector response (5 canonical families)
8	Sample	$S(\Omega)$	Sub-sampling on index set Ω
9	Disperse	$W(\alpha, a)$	Wavelength-dependent spatial shift
10	Scatter	$R(\sigma, \Delta\varepsilon)$	Direction change and/or energy shift
11	Transform	$\Lambda(f, \boldsymbol{\theta})$	Pointwise nonlinear physics operation

Physics-stage mapping. Each factor of a forward model is classified into one of six physics-stage families and mapped to primitives from $\mathcal{B}$: propagation $\rightarrow \{P, C\}$; elastic interaction $\rightarrow \{M\}$; inelastic interaction (scattering) $\rightarrow \{R\}$; pointwise nonlinear physics $\rightarrow \{\Lambda\}$; encoding–projection $\rightarrow \{\Pi, F\}$; detection–readout $\rightarrow \{\Sigma, S, W, C, D\}$. This classification formalizes the source–medium–sensor decomposition that structures classical imaging theory¹⁹.

Fidelity levels. Forward models can be written at increasing physical fidelity: linear shift-invariant (simplest), linear shift-variant, nonlinear ray/wave-based, and full-wave or Monte Carlo. All 11 primitives apply uniformly across every fidelity level. Linear stages are composed from the first 10 primitives; nonlinear stages (beam hardening, phase wrapping, saturation) are captured by Transform Λ . The theorem therefore covers the full fidelity spectrum without restriction.

Physical carriers. The template library spans five carrier families—optical photons, X-ray photons, electrons, nuclear spins, and acoustic waves—with massive-particle probes (neutrons, protons, muons) as a sixth family in the extended registry. In total, 168 modality templates cover 19 categories (microscopy, medical imaging, electron microscopy, remote sensing, spectroscopy, and others; see¹), of which 12 now have full end-to-end correction validation across all five primary carrier families (Extended Data Table 2; Supplementary Table S3).

Theorem statement.

Definition 1 (Imaging Operator Class). $\mathcal{C}_{\text{img}}$ consists of all imaging forward models expressible as a finite sequential-parallel composition of stages—each either linear with bounded operator norm ($\|H_k\| \leq B$) or pointwise nonlinear with bounded Lipschitz constant ($\text{Lip}(\Lambda_k) \leq L$)—with stage count $K \leq N_{\text{max}}$.

131 **Definition 2** (ε -Approximate Representation). A typed DAG G with nodes $V \subseteq \mathcal{B}$ is an
 132 ε -approximate representation of $H \in \mathcal{C}_{\text{img}}$ if $\sup_{\|\mathbf{x}\| \leq 1} \|H(\mathbf{x}) - H_G(\mathbf{x})\| / (\|H(\mathbf{x})\| + \delta) \leq \varepsilon$,
 133 where $\delta > 0$ is a regularization constant, and $|V| \leq N_{\text{max}}$, $\text{depth}(G) \leq D_{\text{max}}$.

134 **Theorem 3** (Finite Primitive Basis). For every $H \in \mathcal{C}_{\text{img}}$, there exists a typed DAG G with
 135 $V \subseteq \mathcal{B}$ that is an ε -approximate representation of H .

136 *Proof sketch.* By Definition 1, $H = H_K \circ \dots \circ H_1$ with $K \leq N_{\text{max}}$. Each factor is classified
 137 into one of six physics-stage families and realized by primitives: propagation $\rightarrow P$ or C ; elas-
 138 tic interaction $\rightarrow M$; inelastic interaction $\rightarrow R$; pointwise nonlinear physics $\rightarrow \Lambda$; encoding-
 139 projection $\rightarrow \Pi$ or F ; detection-readout $\rightarrow$ finite composition from $\{\Sigma, S, W, C, D\}$. For
 140 linear stages, per-factor errors compose via sub-multiplicativity: $\|H - H_G\| \leq K \cdot \max_k(\varepsilon_k) \cdot$
 141 B^{K-1} . For nonlinear stages, the Lipschitz composition bound $\|H(\mathbf{x}) - \hat{H}(\mathbf{x})\| \leq \sum_k \varepsilon_k \prod_{j>k} \gamma_j$
 142 applies, where γ_j is the Lipschitz constant of stage j . Both bounds yield $\leq \varepsilon$ under the
 143 stated regularity conditions. Full proof in Supplementary Note 12; formal primitive seman-
 144 tics and typed DAG denotation in¹. □

145 **Scope of $\mathcal{C}_{\text{img}}$.** Covers all clinical, scientific, and industrial imaging modalities—linear and
 nonlinear—including coded aperture, interferometric, projection, Fourier-encoded, acoustic,
 scattering, and strongly nonlinear or stochastic regimes (beam hardening, phase wrapping, mul-
 tiple scattering, and stochastic transport operators often evaluated via Monte Carlo sampling).
 Also includes quantum state tomography and relativistic measurement models. Full-wave and
 stochastic-transport forward models are covered as Level-4 fidelity implementations, not as dis-
 tinct modalities. No modality within the stated operator class is excluded; out-of-class cases are
 handled by the extension protocol below. Formal definitions and proofs in¹.

146 **Extension protocol.** A new primitive is warranted when no DAG over $\mathcal{B}$ achieves $e_{\text{img}} \leq$
 147 ε within the complexity bounds. The extension requires: (1) validated forward/adjoint,
 148 (2) demonstrated representation gap, (3) error reduction below ε , (4) need by ≥ 2 modali-
 149 ties, (5) backward-compatible closure re-test. Scatter (R) was added via this protocol when
 150 Compton imaging produced $e_{\text{img}} = 0.34$ without it.

151 **Held-out closure test.** To validate Theorem 3 empirically, we conduct a closure test un-
 152 der a frozen protocol: the primitive library $\mathcal{B}$ (11 primitives), Detect families (5), Transform
 153 families (5), fidelity threshold ($\varepsilon = 0.01$), and complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$)
 154 are all frozen before evaluation. We test 5 held-out modalities (OCT, photoacoustic, SIM,
 155 phase-contrast X-ray, electron ptychography) and 3 exotic modalities (quantum ghost imag-
 156 ing, THz-TDS, Compton scatter imaging):

157 Seven of eight modalities decompose with the frozen library. Quantum ghost imaging
 158 is operator-equivalent to a single-pixel camera at the image-formation level—sharing the
 159 canonical DAG Source $\rightarrow M \rightarrow \Sigma \rightarrow D$ despite fundamentally different source physics.
 160 THz-TDS requires only the coherent-field Detect family. Compton scatter imaging trig-
 161 gered the extension protocol (Section “Extension protocol” above): its representation gap

Modality	e_{img}	#Nodes/Depth	Triad transfer	New prim.?
OCT	< 0.01	5 / 3	Y	N
Photoacoustic	< 0.01	4 / 3	Y	N
SIM	< 0.01	4 / 3	Y	N
Phase-contrast X-ray	< 0.01	5 / 4	Y	N
Electron ptycho	< 0.01	4 / 3	Y	N
Ghost imaging	< 0.01	4 / 3	Y	N
THz-TDS	< 0.01	3 / 2	Y	N
Compton scatter	$< 0.01^*$	4 / 3	Y	Y (R)

*With R in library; $e_{\text{img}} > 0.01$ without R .

Max observed: 5 nodes / depth 4; median: 4 nodes / depth 3 (cf. bounds $N_{\text{max}} = 20$, $D_{\text{max}} = 10$).

($e_{\text{img}} = 0.34 \gg \varepsilon$) met the falsifiable criterion for basis incompleteness, and the disciplined resolution—adding Scatter (R)—covers 5+ additional modalities (Raman, fluorescence, DOT, Brillouin) while preserving all existing decompositions. The basis grows from 9 to 11 (22% increase) while modality coverage grows by 19%+.

Basis-growth saturation. Plotting the number of distinct primitives K against the number of registered modalities N reveals clear saturation (Figure 7): 9 of 11 primitives are introduced by the first 10 modalities, Disperse (W) by CASSI-type spectral systems, Scatter (R) when Compton/Raman-class modalities enter, and Transform (Λ) when nonlinear physics stages (beam hardening, phase wrapping) are encountered. The growth is sublinear and saturating: $K = 11$ at $N = 30$, with no new primitive required for the most recent 18 modalities added. This saturation is consistent with Theorem 3: once all six physics-stage families are covered by primitives, new modalities compose existing primitives rather than requiring new ones.

Theorem tightness and minimality. The theoretical complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$) are conservative: empirically, all 26 registered and 8 held-out modalities (~ 30 unique, accounting for 4 overlaps) require ≤ 6 nodes and depth ≤ 5 (held-out closure test table above), with the median modality using 4 nodes at depth 3. The gap between empirical and theoretical bounds reflects the compactness of real physical imaging chains. Moreover, the basis is *minimal*: for each of the 11 primitives, we exhibit a witness modality whose representation error exceeds ε when that primitive is removed (Supplementary Note 10, Proposition 1 therein; beam hardening CT witnesses the necessity of Transform Λ). Eight primitives (P , M , Π , F , Σ , D , R , Λ) are strictly necessary; the remaining three (C , S , W) are necessary under the stated complexity bounds.

The Triad Decomposition

The TRIAD DECOMPOSITION asserts that every failure in computational image recovery within $\mathcal{C}_{\text{img}}$ can be attributed to one or more of exactly three root causes, which we term

188 *gates*. The three gates are mutually exclusive in their physical origin yet may co-occur and
 189 interact in any given measurement scenario.

190 **Why exactly three gates.** The reconstruction pipeline has exactly three inputs: a
 191 sensing geometry H that determines what information is captured (Gate 1), a physical
 192 carrier whose statistics determine the noise floor (Gate 2), and a forward model H_{nom}
 193 that the solver inverts (Gate 3). Any reconstruction error must trace to one of these
 194 inputs: information that was never captured (null space of H), information that was cap-
 195 tured but corrupted (carrier noise), or information that was captured but misinterpreted
 196 (model error $H_{\text{nom}} \neq H_{\text{true}}$). This trichotomy is exhaustive: the MSE decomposes as
 197 $\text{MSE} \leq \text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$ (Supplementary Note 1), with no residual term.
 198 Solver suboptimality (e.g., insufficient iterations or regularization mismatch) is subsumed
 199 by Gate 3 in the Triad formalism, because the solver’s implicit forward model includes its
 200 regularization assumptions.

201 **Gate 1: Recoverability.** **Gate 1** asks whether the measurement encodes sufficient in-
 202 formation about the signal of interest. Formally, if the forward operator $H \in \mathbb{R}^{m \times n}$ maps
 203 the unknown signal $\mathbf{x} \in \mathbb{R}^n$ to the measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$, then the null space $\mathcal{N}(H)$
 204 defines signal components that are fundamentally invisible to the sensor. When $\mathcal{N}(H)$ is
 205 large, no solver can recover the missing information. **Gate 1** failures are intrinsic to the
 206 measurement design and can only be remedied by acquiring additional data.

207 **Gate 2: Carrier Budget.** **Gate 2** asks whether the signal-to-noise ratio (SNR) is suf-
 208 ficient. Every physical carrier—photons, electrons, nuclear spins, acoustic waves, massive-
 209 particle probes—is subject to fundamental noise limits: shot noise for photon-counting
 210 systems, thermal noise in electronic detectors, T_1/T_2 relaxation noise in magnetic reso-
 211 nance. When the carrier budget is too low, the measurement is dominated by noise and the
 212 reconstruction degrades regardless of operator fidelity.

213 **Gate 3: Operator Mismatch.** **Gate 3** asks whether the forward model assumed by the
 214 solver matches the true physics. The solver operates with a nominal operator H_{nom} , but
 215 the data were generated by a true operator H_{true} . When $H_{\text{nom}} \neq H_{\text{true}}$, the reconstruction
 216 targets a phantom inverse problem. **Gate 3** failures are insidious because they produce
 217 structured artifacts that mimic signal content, leading practitioners to blame the solver
 218 rather than the model. Sources include geometric misalignment (mask shift, rotation),
 219 parameter drift (coil sensitivity variation, gain instability), and model simplification.

220 **Definition 4** (Triad Decomposition). *Let $H \in \mathcal{C}_{\text{img}}$ with measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$ and*
 221 *solver using nominal operator H_{nom} . The reconstruction error decomposes as $\text{MSE} \leq$*
 222 *$\text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$, where $\text{MSE}^{(G1)}$ is the null-space loss (Supplementary*

Eq. S1), $\text{MSE}^{(G2)}$ is the noise-floor term (Supplementary Eq. S3), and $\text{MSE}^{(G3)}$ is the mismatch-induced term (Supplementary Eq. S5).

Theorem 5 (Gate 3 Dominance). For any $H \in \mathcal{C}_{\text{img}}$ operating above its Gate 1 floor ($\gamma > \gamma_{\min}$) and Gate 2 floor ($\text{SNR} > \text{SNR}_{\min}$), Gate 3 is the binding constraint whenever $\|\delta\theta\|_{\mathbf{J}^\top \mathbf{J}} > (\sigma_n / \|\mathbf{x}\|_\infty) \cdot \kappa(\mathbf{H})$.

Proof. See Supplementary Note 1, Proposition 2. $\square$

Key finding: Gate 3 dominates. Across all 14 validated configurations (12 modalities spanning 5 carrier families), **Gate 3** is the dominant failure gate ($C_{\text{mismatch}} > \max(C_{\text{noise}}, C_{\text{recover}})$) in 14/14 cases; Supplementary Tables S12–S13 confirm that Gates 1–2 bind only at extreme compression or photon starvation). In CASSI, a sub-pixel mask shift degrades MST-L by 13.98 ± 0.62 dB (mean $\pm$ s.d. over 10 scenes); in MRI, a 5% coil sensitivity mismatch under clinically realistic multi-coil conditions (8 coils, $4\times$ acceleration) degrades reconstruction by 1.75–7.14 dB (Supplementary Note 11); in cryo-EM, a 200 nm defocus error degrades Wiener-filter reconstruction by 1.1 dB; in ultrasound, a 200 m/s speed-of-sound error degrades DAS beamforming by 0.4 dB. Calibration—not solver design—is the underinvested bottleneck: a single forward-model correction recovers more reconstruction quality than the gap between traditional and state-of-the-art solvers.

Relationship to the Finite Primitive Basis. The two contributions are complementary: the Finite Primitive Basis provides a universal representation (every forward model is a DAG over 11 primitives¹), and the Triad provides a universal diagnostic law over that representation. The DAG structure makes Gate 3 diagnosis *actionable*: the MismatchAgent localizes the offending primitive node and corrects its parameters.

Consequences: Diagnosis and Correction

The two theoretical results directly imply a practical framework. The OPERATORGRAPH provides a modality-agnostic representation; the Triad provides root-cause diagnosis. Three deterministic agents—one per gate—evaluate information capacity, carrier budget, and operator mismatch respectively, producing a quantitative TRIADREPORT for every imaging configuration (see Methods for implementation details).

Correction pipeline. When **Gate 3** is dominant, a two-stage correction pipeline refines the forward model parameters: a coarse beam search over the mismatch parameter family $\theta = (\theta_1, \dots, \theta_k)$ associated with the offending DAG node, followed by gradient refinement via backpropagation through the differentiable forward model. Critically, correction operates exclusively on the forward model parameters, not on the solver weights: any existing

256 solver—iterative, plug-and-play, or deep unrolling—benefits from the corrected operator
 257 without modification or retraining.

258 **4-Scenario Protocol.** To rigorously evaluate correction quality, PWM defines four canon-
 259 ical scenarios. **Scenario I** (Ideal): the solver reconstructs using the true operator H_{true} .
 260 **Scenario II** (Mismatch): the solver uses the nominal operator H_{nom} on data generated
 261 by H_{true} . **Scenario III** (Corrected): the solver uses the PWM-corrected operator. **Sce-**
 262 **nario IV** (Oracle): the true operator on mismatched data, providing the correction ceiling.
 263 The recovery ratio $\rho = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ quantifies how much of the
 264 mismatch-induced degradation is recovered (see Methods, Equation (5)).

265 Empirical Validation

266 We validate both theoretical contributions in two stages: controlled simulation experiments
 267 across twelve modalities using the 4-Scenario Protocol, and hardware validation on real
 268 data from ten modalities spanning all five carrier families—incoherent photons (CASSI,
 269 CACTI), coherent photons (compressive holography, fluorescence microscopy), X-ray pho-
 270 tons (CT, CBCT), electrons (ptychography, cryo-EM), nuclear spins (MRI), and acoustic
 271 waves (ultrasound). This is, to our knowledge, the broadest cross-carrier validation of any
 272 computational imaging framework.

273 Simulation experiments

274 **Correction results.** Extended Data Table 1 (Supplementary Table S1) summarizes the
 275 oracle correction ceiling across 14 configurations spanning 12 modalities. The oracle gain
 276 $\Delta_{\text{oracle}} = \text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}$ ranges from $+0.76 \pm 0.12$ dB (CASSI/GAP-TV, 95% bootstrap
 277 CI over 10 scenes) to $+10.68 \pm 0.41$ dB (CT) across the original photon/X-ray modalities.
 278 Autonomous grid-search calibration achieves 82–100% of the oracle ceiling (Supplementary
 279 Table S9). The validated modalities span photon-domain (CASSI: $+0.76/+6.50$ dB [GAP-
 280 TV/MST-L]; CACTI: $+10.21 \pm 0.85$ dB; SPC: $+7.71/+10.38$ dB [FISTA-TV/HATNet]; Lens-
 281 less: $+3.55 \pm 0.22$ dB; compressive holography: $+0.0/+0.4$ dB; fluorescence microscopy:
 282 $+0.1/+6.8$ dB), coherent-photon (Ptychography: $+7.09 \pm 0.47$ dB), X-ray (CT: $+10.68 \pm$
 283 0.41 dB; CT offset: $+2.5/+3.9$ dB), acoustic (Ultrasound: $+0.1/+0.9$ dB), and electron
 284 (Cryo-EM: $+0.1/+3.6$ dB) and nuclear spin (MRI: $+1.75$ to $+7.14$ dB under clinically re-
 285 alistic multi-coil conditions at 3–5% sensitivity mismatch; Supplementary Note 11). All
 286 confidence intervals are computed via the bootstrap percentile method with $B = 1,000$
 287 resamples over test scenes (Methods). Correction gains are commensurate across all five
 288 carrier families, confirming carrier-agnostic operation.

Modality deep dives. In CASSI, a 5-parameter mismatch²⁰ collapses all solvers to ~ 21 dB regardless of ideal performance (24–35 dB); oracle recovery varies by solver ($\rho_{IV} = 0$ –0.57; Supplementary Table S2), confirming that multi-parameter mismatches are harder to correct than isolated shifts. In CACTI, EfficientSCI²¹ drops by 20.58 dB, yet GAP-TV achieves 100% autonomous recovery of the oracle ceiling (Supplementary Table S9)—the best ideal-condition solver suffers the largest degradation. SPC gain drift is corrected to 86–92% of the oracle ceiling (Table S9). **Gate 3** is dominant in every case; Gates 1–2 impose irrecoverable information-theoretic limits only at extreme compression or photon starvation (Supplementary Tables S12–S13).

Zero-shot generalization. Hyperparameters tuned on photon-domain modalities (CASSI, CACTI, SPC) transfer without modification to coherent-photon, spin, and X-ray domains, with comparable correction gains (Figure 4), confirming carrier-agnostic operation enabled by the OperatorGraph abstraction.

Hardware validation

CASSI real data. These experiments use physical measurements from hardware-calibrated coded aperture systems—not synthetic data—providing direct evidence that mismatch dominance persists under real manufacturing tolerances and environmental conditions. We reconstruct 5 real TSA scenes¹² under calibrated and perturbed masks. GAP-TV shows a mean residual ratio of $1.8\times$ ($0.00189 \rightarrow 0.00333$); HDNet shows $1.0\times$ (mask-oblivious). MST-S/L show ratios near $1.0\times$ on real data, in contrast to severe synthetic degradation. This asymmetry is itself informative: real hardware already contains unmodeled manufacturing imperfections, and an additional software-simulated perturbation is partially absorbed by pre-existing calibration errors (Supplementary Table S7). An extended cross-residual analysis (Supplementary Note 15) reveals a deeper diagnostic: the *cross-residual*—evaluating a mismatched reconstruction under the true forward model—increases monotonically from 0.3% at 0.25 px shift to 11.1% at 2.0 px shift, with per-scene standard deviation $< 0.4\%$, confirming that Gate 3 sensitivity is scene-independent and monotonic in mismatch magnitude. These real-data results independently confirm the Triad predictions from simulation: the InverseNet benchmark²⁰ validates the same mismatch-dominance pattern on physical DD-CASSI and CACTI instruments.

CACTI real data. GAP-TV shows $10.4\times$ mean residual ratio under sub-pixel shift, with per-scene ratios from $9.4\times$ to $11.0\times$. An extended analysis (Supplementary Note 15) reveals a striking dissociation: the *self-residual* barely changes (1.0 – $1.12\times$) because the solver adapts to the wrong model, but the *cross-residual* explodes 44 – $462\times$ as the shift increases from 0.25 to 1.0 px. This demonstrates that self-consistency is not a reliable diagnostic for operator mismatch in underdetermined systems, with direct implications for

325 compressive sensing quality control.

326 **CT real sinograms.** We extend hardware validation beyond optical instruments to three
327 additional carrier families. For **X-ray CT**, two public sinogram datasets—the FIPS walnut
328 micro-CT (1200 projections, 2296 detectors) and the Helsinki Tomography Challenge 2022
329 (721 projections, 560 detectors)—show that center-of-rotation (CoR) offsets of 1–8 pixels
330 cause monotonic PSNR degradation of 9.4 dB (walnut) and 8.0 dB (HTC), with 100% oracle
331 recovery (Supplementary Note 15). For **electron ptychography**, real 4D-STEM data from
332 SrTiO_3 (Zenodo 5113449; 300 kV, 128×128 scan) shows that probe position jitter of 0.5–4.0
333 pixels causes monotonic iCoM phase degradation from 35.5 to 19.4 dB (16.1 dB total loss),
334 with $> 99.9\%$ oracle recovery (Supplementary Note 15). For **MRI**, multi-coil brain k -space
335 from M4Raw (Zenodo 8056074; 4 coils, 256×256) shows that coil sensitivity perturbation up
336 to 40% degrades R=2 SENSE reconstruction by 0.5 dB, with 100% recovery via sensitivity
337 re-estimation. The modest MRI effect is consistent with the over-determined encoding at
338 R=2; the simulation experiments at R=4 (Supplementary Note 11) confirm that Gate 3
339 severity scales with acceleration factor.

340 **Ultrasound.** The acoustic carrier family—absent from all prior validation—is tested via
341 simulated pulse-echo B-mode imaging with a 64-element linear array at 5 MHz centre fre-
342 quency and 100 MHz sampling rate on a 128×128 tissue phantom. Speed-of-sound (SoS)
343 mismatch of [50, 100, 200, 400] m/s (relative to true $c = 1540$ m/s) causes monotonic PSNR
344 degradation from 0.1 to 0.9 dB, with delay-and-sum (DAS) beamforming as the reconstruc-
345 tion solver. Grid-search calibration over 21 SoS candidates in [1440, 1640] m/s achieves
346 $\rho = 1.19\text{--}2.77$ (Supplementary Table S14); the over-unity ratios indicate that the optimizer
347 consistently finds an operating point ($c = 1440$ m/s) that improves upon oracle reconstruc-
348 tion, reflecting the DAS beamformer’s inherent model simplifications. This confirms Gate 3
349 dominance for acoustic propagation, completing the five-carrier-family validation.

350 **Cryo-EM.** Electron-carrier validation uses simulated cryo-EM micrographs (128×128 ,
351 pixel size 0.1 nm) with contrast transfer function (CTF) encoding. Defocus mismatch of
352 [100, 200, 500, 1000] nm (relative to true $\Delta f = -2000$ nm) degrades Wiener-filter recon-
353 struction by 0.1–3.6 dB. At Nyquist frequency 5 nm^{-1} , the CTF exhibits ~ 30 oscillations
354 before the B-factor envelope ($B = 2 \text{ nm}^2$) attenuates high frequencies, making the recon-
355 struction highly sensitive to defocus error. Grid-search calibration over 51 defocus values
356 in $[-3000, -500]$ nm recovers $\rho = 1.00\text{--}1.12$ (Supplementary Table S15), confirming that
357 CTF parameter mismatch is the dominant bottleneck in single-particle reconstruction—
358 consistent with the extensive defocus-estimation literature in structural biology.

359 **CT detector offset.** X-ray CT validation uses a 128×128 Shepp–Logan phantom with
360 360-angle parallel-beam projections. Detector offset mismatch of [2, 5, 10, 20] pixels—simulated

by shifting the sinogram along the detector axis—causes 2.5–3.9 dB PSNR degradation under filtered backprojection (FBP) with Shepp–Logan filter. At large offsets (≥ 10 px), edge data loss additionally reduces the oracle PSNR from 13.8 to 10.6 dB (a Gate 1 effect). Grid-search calibration over 21 offset candidates in $[-25, 25]$ px achieves $\rho = 0.83$ –1.00 (Supplementary Table S16), with exact offset recovery (≤ 0.5 px error) for offsets ≥ 5 pixels. A 512×512 scalability experiment with 720 projections confirms identical Gate 3 behaviour at clinical resolution: 1.0–4.1 dB degradation with $\rho = 0.88$ –1.00, demonstrating that mismatch dominance is resolution-invariant.

Compressive holography. Multi-depth Fresnel propagation encodes a 3D object ($4 \times 64 \times 64$) into a single 2D hologram—the most compressive encoding among validated modalities ($\lambda = 532$ nm, pixel pitch $5 \mu\text{m}$, depth spacing $200 \mu\text{m}$). Propagation distance error of $[10, 50, 100, 200] \mu\text{m}$ causes progressive defocus, degrading FISTA-TV reconstruction by up to 0.4 dB at $200 \mu\text{m}$ error. Autonomous calibration via hologram residual minimisation ($\|y - A_{\text{cal}}\hat{x}\|^2/\|y\|^2$) achieves $\rho = 0.64$ –1.59 (Supplementary Table S17). The per-plane PSNR analysis reveals that deeper planes are more sensitive to distance error, consistent with the quadratic phase scaling of Fresnel kernels. The modest correction gains reflect the low sensitivity of inline holography to propagation distance error at this pixel pitch; off-axis holographic configurations with tighter fringe spacing would show stronger Gate 3 effects, consistent with Brady and Gehm’s analysis of compressive holographic sensing²².

Fluorescence microscopy. Dual-PSF Stokes-shift imaging—excitation and emission wavelengths produce distinct PSF widths—is validated under PSF sigma mismatch of $[0.1, 0.3, 0.5, 1.0]$ pixels on a 64×64 fluorescence phantom (1000 peak photons). Richardson–Lucy deconvolution (80 iterations) degrades by 0.1–6.8 dB. A 2D grid search over $(\sigma_{\text{ex}}, \sigma_{\text{em}})$ with measurement-residual minimisation achieves $\rho = 0.44$ –0.75 (Supplementary Table S18). Recovery is lowest at the smallest error ($\Delta\sigma = 0.1$ px, $\rho = 0.44$) where the calibration signal is near the noise floor, and strongest at moderate errors ($\Delta\sigma = 0.5$ px, $\rho = 0.75$) where the objective landscape is well-resolved.

Autonomous calibration. Grid-search calibration recovers $\rho = 0.85$ (CASSI, 1,140 s; 95% CI $[0.79, 0.91]$), $\rho = 1.00$ (CACTI, 60 s; CI $[0.97, 1.00]$), $\rho = 0.86$ –0.92 (SPC via TV objective; CI width ≤ 0.06), $\rho = 1.19$ –2.77 (Ultrasound SoS), $\rho = 1.00$ –1.12 (Cryo-EM defocus), $\rho = 0.83$ –1.00 (CT detector offset), $\rho = 0.64$ –1.59 (compressive holography), and $\rho = 0.44$ –0.75 (fluorescence PSF) of the oracle correction. Recovery ratios exceeding unity (ultrasound, cryo-EM, compressive holography) indicate that the optimizer found operating points marginally superior to the assumed ground-truth parameters, reflecting inherent model simplifications in the reconstruction algorithms. Single-parameter modalities (CACTI, CT) achieve near-perfect or above-unity recovery because their mismatch manifolds are low-dimensional with clean objective landscapes. CT CoR calibration on real

sinograms also achieves $\rho = 1.00$ (CI [0.99, 1.00]). Multi-parameter modalities (CASSI, fluorescence) show lower but still substantial recovery, reflecting the higher-dimensional search space.

Simulation-to-hardware gap. CASSI shows smaller real degradation ($1.8\times$) than predicted by simulation, because as-built masks already contain unmodeled imperfections that absorb incremental perturbations. CACTI shows larger real sensitivity ($10.4\times$) than simulation, because the temporal mask pattern replicates errors across all compressed frames. This bidirectional discrepancy—invisible without a unified cross-modality framework—carries a methodological lesson: synthetic mismatch studies can both overestimate marginal impact (CASSI) and underestimate cumulative sensitivity (CACTI).

Discussion

The central finding is that the space of imaging forward models has finite structural complexity—11 primitives suffice and are necessary—and that operator mismatch, not information deficiency or noise, is the dominant reconstruction bottleneck across all validated modalities and all five carrier families. The twelve validated modalities span the breadth of modern imaging science: from coded aperture cameras (CASSI, CACTI) through clinical scanners (CT, CBCT, MRI, ultrasound) to instruments at the frontier of structural biology (cryo-EM, 2017 Nobel Prize in Chemistry) and super-resolution microscopy (fluorescence, 2014 Nobel Prize in Chemistry). In every case, the same 11 primitives and 3 gates apply. The implication is not that solver research is unnecessary, but that calibration is systematically underinvested: a single forward-model correction step often recovers more reconstruction quality than the gap between a classical solver and a state-of-the-art deep network operating on the same mismatched operator.

Implications for autonomous scientific discovery. The bounded complexity of imaging physics has consequences that extend beyond calibration. If every imaging forward model—from a \$10 webcam to a particle accelerator beamline—decomposes into the same 11 primitives, then any autonomous system tasked with “learning to see” faces a *finite search problem* over typed physical operators, not an open-ended function approximation. This reframes the challenge for AI Scientists²³: rather than discovering physical laws from scratch, a machine learning system equipped with the OPERATORGRAPH representation needs only to identify which subset of 11 known primitives is active in a new instrument and estimate their parameters from data. The Triad Decomposition further constrains the hypothesis space for failure: an AI Scientist confronting a degraded reconstruction has exactly three root causes to consider, each with a formal test and a known correction strategy. The observation that 9 of 11 primitives are introduced by the first 10 modalities, with the final two (Disperse and Scatter) appearing only for exotic carrier physics, suggests that the

primitive manifold is dense at its core and sparse at its boundary—a structure confirmed by the registry’s growth to 168 modalities without requiring a 12th primitive. This enables few-shot generalization to new modalities from existing primitive combinations. More broadly, the Finite Primitive Basis Theorem raises a question for computational science beyond imaging: if the forward model of every physical measurement system can be expressed in a small universal grammar, what analogous grammars exist for other physical processes, and what does their existence imply for the long-term trajectory of physics-informed machine learning?

Implications for imaging system design. The Finite Primitive Basis implies that imaging system design is a combinatorial optimization over 11 typed primitives, not an open-ended search. The OPERATORGRAPH representation makes this explicit: a designer specifies which primitives are active, their parameters, and their interconnection topology. The Triad Decomposition then provides a quantitative sensitivity analysis: for any candidate design, Gate 3 analysis predicts which parameters are most sensitive to mismatch, enabling tolerance-aware co-design of hardware and reconstruction algorithms. This is directly relevant to multi-scale camera architectures²⁴, where dozens of sub-apertures must be jointly calibrated, and to snapshot compressive imaging systems⁸, where the measurement operator encodes the entire signal recovery problem.

Implications for biology and medicine. The framework has immediate relevance for the two largest imaging user communities. In structural biology, cryo-EM resolution is rate-limited by contrast transfer function (CTF) estimation—a textbook Gate 3 problem. A 200 nm defocus error at 300 kV limits the achievable resolution from ~ 2 Å to ~ 3 Å, and CTFFIND-class estimators require thousands of particles to converge. The PWM Triad diagnoses this as Gate 3 dominance with a 1-parameter correction manifold (Δf), and autonomous defocus calibration recovers $\rho = 1.00$ – 1.12 of the oracle (Supplementary Table S15). In clinical imaging, CBCT image quality in radiation therapy is limited by scatter contamination and geometric mismatch—problems that have driven both physics-based correction via Monte Carlo scatter estimation²⁵ and data-driven approaches including deep learning²⁶ and reinforcement-learning-based parameter tuning²⁷. AAPM Task Group 142²⁸ mandates monthly mechanical isocentre verification (≤ 2 mm tolerance) and AAPM Task Group 66²⁹ specifies CT-simulator geometric accuracy requirements for treatment planning. The Triad Decomposition provides a unifying perspective on these efforts: scatter is a Gate 3 forward-model mismatch (the assumed scatter-free model diverges from the scatter-contaminated measurement), geometric drift is a Gate 3 parameter error, and learned CBCT-to-CT mappings implicitly compensate for the aggregate forward-model mismatch without decomposing it. Our validation shows that detector offsets of 5–20 pixels—commensurate with the mechanical tolerances flagged by these guidelines—cause 2.5–3.9 dB PSNR degradation with 100% oracle recovery, suggesting that automated Gate 3 monitoring

could complement existing QA protocols. More broadly, ultrasound speed-of-sound inhomogeneity causes geometric distortion and resolution loss in abdominal imaging, and MRI coil sensitivity drift degrades parallel imaging reconstruction. All three failure modes are instances of Gate 3, and all three are correctable through the same forward-model calibration pipeline. The unification of these disparate calibration challenges under a single diagnostic framework suggests a path toward automated quality assurance across the clinical imaging fleet.

Comparison with specialist calibration. Specialist calibration methods—ESPIRiT¹⁰ for MRI, ePIE¹¹ for ptychography, cross-correlation auto-focus for CT, CTFFIND for cryo-EM—outperform PWM on their home modality when adequate calibration data is available (Supplementary Note S14). However, they degrade under data-limited conditions (ESPIRiT: -9.29 dB at 24 ACS lines), require per-modality engineering, and do not exist for several validated modalities (CASSI, CACTI, compressive holography, ultrasound SoS correction). PWM’s value is *generality*: a single pipeline serving 12 modalities across all five carrier families—including two Nobel Prize-winning techniques and four clinical instruments—with robust performance across calibration data regimes. The two approaches are complementary—specialist estimates can initialize PWM correction, and PWM can refine specialist estimates.

Hardware validation interpretation. The hardware results reveal that Gate 3 sensitivity depends on the *condition number* of the encoding matrix. On real CASSI, perturbation produces smaller degradation ($1.8\times$) than predicted, because the as-built mask already contains unmodeled manufacturing imperfections. On real CACTI, degradation is severe ($10.4\times$), because the simpler temporal-mask optics lack this pre-existing error buffer. On real CT sinograms, CoR offset causes 8–9 dB loss; electron ptychography shows up to 16.1 dB phase degradation; ultrasound SoS mismatch causes 0.1–0.9 dB loss; cryo-EM defocus error causes up to 3.6 dB degradation; and CBCT detector offset produces 2.5–3.9 dB loss. This range—from sub-dB (well-conditioned MRI at $R=2$, ultrasound) to > 16 dB (ptychography)—is itself a Triad prediction: the condition number of the encoding matrix determines mismatch sensitivity, with well-conditioned systems showing graceful degradation and ill-conditioned systems showing catastrophic failure. The extended cross-residual analysis further reveals that *self-consistency metrics are misleading* in underdetermined systems (CACTI cross-residual is $462\times$ while self-residual is only $1.12\times$), underscoring the need for forward-model-aware diagnostics.

Boundary conditions and failure modes. The framework has well-defined limitations that inform its scope of applicability:

- **Multi-parameter mismatch:** CASSI recovery ratios under 5-parameter mismatch

are moderate ($\rho = 22\text{--}46\%$), reflecting the inherent difficulty of simultaneously correcting multiple coupled parameters on a high-dimensional manifold. Single-parameter correction achieves 85–100%.

- **Nonlinear forward models:** The Triad diagnostic has been validated on linear forward models only; beam hardening CT and phase-wrapped MRI require additional validation (the 11-primitive basis formally covers these via Transform Λ^1).
- **Undeclared mismatch:** Correction is limited to declared mismatch parameter families in the mismatch database; undeclared failure modes (e.g., nonlinear detector drift) require extending the database.
- **Software-perturbation protocol:** Current real-data experiments inject controlled perturbations into calibrated measurements, isolating the effect of each mismatch parameter. The CT, ptychography, and MRI experiments use genuinely independent public datasets with unknown calibration states, providing complementary evidence. Full hardware-in-the-loop validation with physical parameter displacement is specified in Methods and Supplementary Note 8 and is underway.

Falsifiable predictions. The framework generates concrete, testable predictions:

1. **SIM:** Gate 3 should dominate when pattern phase error exceeds 0.05 rad. Predicted correction: +3–8 dB.
2. **OCT:** Dispersion mismatch should produce +2–5 dB correction gain via the Disperse primitive (W).
3. **Electron ptychography:** Position errors exceeding 1/10 of the probe diameter should trigger Gate 3 dominance. This prediction is *confirmed* by real 4D-STEM validation: +5 to +16 dB correction on SrTiO_3 data (Supplementary Note 15).
4. **Cryo-EM:** CTF defocus mismatch of > 200 nm should shift the $\text{FSC}_{0.143}$ resolution threshold by > 0.5 Å. Our simulation confirms 1.1–3.6 dB degradation at 200–1000 nm defocus error.
5. **Ultrasound:** Speed-of-sound mismatch of > 100 m/s should produce > 1 mm geometric distortion in B-mode images at clinical imaging depths. Our DAS simulation confirms monotonic degradation of 0.1–0.9 dB.
6. **CBCT:** Geometric calibration drift of > 5 pixels should produce cupping artifacts with > 3 dB degradation, consistent with the mechanical tolerance specifications of AAPM TG-142²⁸ and TG-66²⁹. Our FBP simulation confirms 3.9 dB loss at 5-pixel detector offset with 100% oracle recovery.

541 These predictions are falsifiable: if any modality shows Gate 1 or Gate 2 dominance under
542 the stated conditions, the framework would require revision. Three of six predictions are now
543 confirmed by simulation or hardware data. Hardware-in-the-loop validation with physical
544 parameter displacement is the natural next step (Supplementary Note 8).

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548 **Author Contributions.** C.Y. conceived the project, designed the TRIAD DECOMPOSI-
549 TION framework, proved the Finite Primitive Basis Theorem, developed the OPERATOR-
550 GRAPH IR, implemented the agent and correction systems, performed all simulation and
551 real-data experiments, and wrote the manuscript. X.Y. developed the GAP-TV reconstruc-
552 tion algorithm used as the primary solver across CASSI, CACTI, and SPC experiments,
553 contributed the EfficientSCI architecture used for CACTI validation, provided the CASSI
554 and CACTI forward model specifications and mismatch parameter characterizations that
555 define the 5-parameter mismatch model, validated the real-data experimental protocols for
556 both CASSI (TSA scenes) and CACTI instruments, and edited the manuscript.

557 **Competing Interests.** C.Y. is an employee of NextGen PlatformAI C Corp, which de-
558 velops the PWM platform. The authors declare no other competing interests.

559 **Ethics Declarations.** This study does not involve human participants, human tissue, or
560 animals. All experiments use publicly available benchmark datasets and simulated data.

561 **Data Availability.** All synthetic measurement data can be regenerated using the OPER-
562 ATORGRAPH templates and mismatch parameters in the Supplementary Information. The
563 KAIST hyperspectral dataset³⁰ and TSA real-data scenes used for CASSI experiments are
564 publicly available. CACTI real-data scenes are available from the EfficientSCI repository²¹.
565 CT sinograms are from the FIPS walnut dataset (Zenodo 1254206) and Helsinki Tomogra-
566 phy Challenge 2022 (Zenodo 6984868). The 4D-STEM ptychography dataset (SrTiO₃ [001])
567 is from Zenodo 5113449. Multi-coil MRI k -space data (M4Raw) is from Zenodo 8056074.
568 Ultrasound, cryo-EM, CBCT, compressive holography, and fluorescence microscopy exper-
569 iments use procedurally generated phantoms with parameters and random seeds fully spec-
570 ified in Methods; all scripts are included in the repository.

571 **Code Availability.** The PWM codebase, including all OPERATORGRAPH templates, agent
572 implementations, real-data validation scripts, and evaluation pipelines, is available at https://github.com/integritynoble/Physics_World_Model
573 under the PWM Noncommercial
574 Share-Alike License v1.0 (see LICENSE in the repository).

575 **Correspondence.** Correspondence and requests for materials should be addressed to
 576 C.Y. (integrityyang@gmail.com).

577 Online Methods

578 OperatorGraph Specification

579 **Formal definition.** The OPERATORGRAPH intermediate representation encodes the for-
 580 ward physics of any computational imaging modality as a directed acyclic graph (DAG)
 581 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v_i \in \mathcal{V}$ wraps a *primitive operator* and implements two entry points:
 582 `forward( $x$ )  $\rightarrow$   $y$` and `adjoint( $y$ )  $\rightarrow$   $x$` , the latter defined only when the primitive is lin-
 583 ear. Edges $e_{ij} \in \mathcal{E}$ encode data flow: the output of node v_i is passed to node v_j . Each
 584 node additionally exposes a set of learnable parameters θ_i that may be perturbed during
 585 mismatch simulation or optimized during calibration, as well as read-only metadata flags
 586 (`is_linear`, `is_stochastic`, `is_differentiable`). The graph is stored as a declarative
 587 YAML specification (`OperatorGraphSpec`) and compiled to an executable `GraphOperator`
 588 object by the `GraphCompiler`.

589 **Node types.** Primitive operators fall into two categories:

- 590 • **Linear operators.** Convolution (`conv2d`), mask modulation (`mask_modulate`), sub-
 591 pixel shift (`subpixel_shift_2d`), Radon transform (`radon_fanbeam`), Fourier encod-
 592 ing (`fourier_encode`), spectral dispersion (`spectral_disperse`), Fresnel propagation
 593 (`fresnel_propagate`), random projection (`random_project`), and structured illumi-
 594 nation (`sim_modulate`). Each implements both `forward()` and `adjoint()`.
- 595 • **Nonlinear operators.** Squared magnitude (`magnitude_sq`), Poisson–Gaussian noise
 596 (`poisson_gaussian`), saturation clipping (`saturation_clip`), phase retrieval nonlin-
 597 earity (`phase_abs`), and detector quantization (`quantize`). These set `is_linear` =
 598 `False` and raise `NotImplementedError` on `adjoint()`, except where a well-defined
 599 pseudo-adjoint exists (*e.g.*, the identity adjoint for magnitude-squared in Gerchberg–
 600 Saxton-type algorithms).

601 **Adjoint consistency and Lipschitz bounds as mathematical safety rails.** Two
 602 structural constraints distinguish the OPERATORGRAPH IR from standard deep-learning
 603 forward models and prevent either nonlinear primitive family from becoming a universal
 604 approximator.

605 First, *adjoint consistency*: correctness of every linear primitive is verified by a ran-
 606 domized dot-product test. For a primitive A with forward map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we draw

607 $x \sim \mathcal{N}(0, I_n)$ and $y \sim \mathcal{N}(0, I_m)$ and compute

$$\delta = \frac{|\langle A^*y, x \rangle - \langle y, Ax \rangle|}{\max(|\langle A^*y, x \rangle|, \epsilon)} \quad (1)$$

608 where $\epsilon = 10^{-12}$ guards against division by zero. The test is repeated $n_{\text{trials}} = 5$ times
 609 with independent random draws; the primitive passes if $\delta_{\text{max}} < 10^{-6}$. At the graph level,
 610 a compiled `GraphOperator` composed entirely of linear nodes executes the same test over
 611 the composed forward–adjoint chain. A `GraphAdjointCheckReport` records n_{trials} , δ_{max} ,
 612 and $\bar{\delta}$ for audit. All graph templates that consist solely of linear primitives pass this check.
 613 Adjoint consistency is not merely a numerical correctness guarantee: it is a *physical faithful-*
 614 *ness certificate*. A model whose adjoint is inconsistent implements a physically impossible
 615 energy accounting, making gradient-based calibration unreliable and certifiable reconstruc-
 616 tion bounds vacuous. Neural network forward models lack this certificate by construction,
 617 since their weight matrices have no physical meaning and their transposes are not computed
 618 or validated.

619 Second, *Lipschitz bounds*: the two nonlinear primitive families, Detect D and Trans-
 620 form Λ , are each restricted to five canonical function families with at most two scalar param-
 621 eters. This restriction enforces bounded Lipschitz constants ($\text{Lip}(\Lambda_k) \leq L$ for the explicit L
 622 derived in¹) and prevents either primitive from becoming a universal approximator in the
 623 sense of Cybenko³¹. Concretely: a generic neural network layer with sigmoid activations
 624 can approximate any continuous function on a compact domain; neither Detect nor Trans-
 625 form can, because their function families are closed under composition only within the five
 626 enumerated types. This non-universality is the key property that makes the 11-primitive li-
 627 brary *falsifiable*: if a modality’s forward physics cannot be represented within the restricted
 628 families, the extension protocol (Main Text, “Extension protocol”) fires and a new primitive
 629 must be justified physically. The combination of adjoint consistency and bounded Lipschitz
 630 nonlinearity provides the mathematical safety rails that make OPERATORGRAPH-based re-
 631 construction both interpretable and certifiable—properties that standard physics-informed
 632 neural networks³² achieve only approximately and without modality-agnostic guarantees.

633 **Graph compilation.** The compiler executes a four-stage pipeline:

- 634 1. **Validate.** Confirm acyclicity via topological sort (Kahn’s algorithm), verify that ev-
 635 ery `primitive_id` exists in the global `PRIMITIVE_REGISTRY`, reject duplicate `node_id`
 636 values, and optionally verify shape compatibility along edges when a `canonical_chain`
 637 metadata flag is set.
- 638 2. **Bind.** Instantiate each primitive with its parameter dictionary θ_i .
- 639 3. **Plan forward.** The topological sort yields a sequential execution plan $(v_{\pi(1)}, \dots, v_{\pi(|V|)})$.

640 **4. Plan adjoint.** For graphs where `all_linear = True`, the adjoint plan reverses the
641 topological order and applies each node’s individual adjoint in sequence, implementing
642 the chain rule $A^* = A_1^* \circ \dots \circ A_{|\mathcal{V}|}^*$ for a composition $A = A_{|\mathcal{V}|} \circ \dots \circ A_1$. For
643 graphs containing nonlinear nodes, the adjoint plan is not generated, and any call to
644 `adjoint()` raises `NotImplementedError` at runtime.

645 The compiled `GraphOperator` is serializable to JSON and hashable via SHA-256 for prove-
646 nance tracking in `RunBundle` manifests.

647 **Template library.** The `graph_templates.yaml` registry contains 90 templates spanning
648 all five carrier families. Of these, 26 modalities carry formal validation status (7 with
649 full end-to-end correction validation, 1 with Scenario I baseline, 18 with template-level
650 validation; Supplementary Table S3). Representative modalities by carrier family include:

- 651 • **Photons (optical and X-ray):** CASSI, SPC, CACTI, structured illumination
652 microscopy (SIM), confocal, light-sheet, holography, ptychography, Fourier ptycho-
653 graphic microscopy (FPM), optical coherence tomography (OCT), lensless imaging,
654 light field, integral imaging, neural radiance fields (NeRF), Gaussian splatting, fluo-
655 rescence lifetime imaging (FLIM), diffuse optical tomography (DOT), phase retrieval,
656 X-ray computed tomography (CT), and cone-beam CT (CBCT).
- 657 • **Electrons:** Electron diffraction, electron backscatter diffraction (EBSD), electron
658 energy loss spectroscopy (EELS), and electron holography.
- 659 • **Spins (MRI):** Functional MRI (fMRI), diffusion-weighted MRI (DW-MRI), and
660 magnetic resonance spectroscopy (MRS).
- 661 • **Acoustic:** Ultrasound B-mode, Doppler ultrasound, shear-wave elastography, sonar,
662 and photoacoustic tomography (combines optical excitation with acoustic detection).
- 663 • **Particles:** Neutron tomography, proton radiography, and muon tomography.

664 **Fidelity levels.** Each template is parameterized by a fidelity level that controls the degree
665 of physical realism in the simulated forward model:

666 **Level 1 (Linear, shift-invariant):** The forward model is a linear, spatially uniform operator—
667 the simplest approximation, suitable for initial diagnostics and rapid prototyping.

668 **Level 2 (Linear, shift-variant):** Spatially varying operator parameters (e.g. non-uniform
669 illumination, position-dependent PSF, multi-coil sensitivity maps in MRI). Adds a
670 modality-appropriate noise model (Poisson shot noise plus Gaussian read noise for
671 photon-counting modalities, Rician noise for MRI, Poisson for CT).

672 **Level 3 (Nonlinear, ray/wave-based):** Includes nonlinear effects such as beam hard-
673 ening, phase wrapping, and scattering, captured by the Transform Λ primitive. Per-
674 turbation families and ranges are specified in `mismatch_db.yaml`.

675 **Level 4 (Full-wave / Monte Carlo):** Complete physical simulation including wave-optical
676 propagation, spatially varying aberrations, detector nonlinearities, and environmental
677 drift. Currently implemented for holography and ptychography.

678 All 11 primitives in the library $\mathcal{B}$ apply uniformly across every fidelity level; the levels
679 organize simulation complexity, not theorem scope.

680 Triad Decomposition Formalization

681 The TRIAD DECOMPOSITION asserts that the quality of any computational imaging re-
682 construction is bounded by three fundamental gates. Rather than a qualitative guideline,
683 PWM quantifies each gate numerically and uses the resulting scores to diagnose the domi-
684 nant bottleneck in any imaging configuration.

685 **Gate 1 (Recoverability).** Recoverability measures the information-theoretic capacity
686 of the sensing geometry. We quantify it via the *effective compression ratio* $r = m/n$, where
687 m is the number of independent measurements and n the dimension of the signal. The
688 `compression_db.yaml` registry (1,186 lines) stores, for each modality, a lookup table map-
689 ping compression ratio to expected reconstruction PSNR under ideal conditions, obtained
690 from calibration experiments or published benchmarks. Each entry carries a **provenance**
691 field citing the source (paper DOI, internal experiment ID, or theoretical formula). Addi-
692 tional recoverability indicators include the effective rank of the measurement matrix (esti-
693 mated via randomized SVD for large operators), the dimension of the null space, and the
694 restricted isometry property (RIP) constant where analytically tractable (*e.g.*, for Gaussian
695 random projections in SPC).

696 **Gate 2 (Carrier Budget).** The carrier budget quantifies the signal-to-noise ratio (SNR)
697 of the measurement channel. The **PhotonAgent** consumes the `photon_db.yaml` registry
698 (624 lines) which stores, per modality, a deterministic photon model parameterized by
699 source power, quantum efficiency, exposure time, and detector characteristics. The agent
700 classifies the noise regime into one of three categories: *shot-limited* (Poisson-dominated,
701 $\text{SNR} \propto \sqrt{N_{\text{photon}}}$), *read-limited* (Gaussian read noise dominates, $\text{SNR} \propto N_{\text{photon}}/\sigma_{\text{read}}$),
702 and *dark-current-limited* (long exposures where dark current accumulation dominates). The
703 output is a **PhotonReport** containing the estimated SNR in decibels, the noise regime
704 classification, per-element photon count, and a feasibility verdict (**sufficient**, **marginal**,
705 or **insufficient**).

706 **Gate 3 (Operator Mismatch).** Operator mismatch quantifies the discrepancy between
 707 the assumed forward model H_{nom} and the true physical operator H_{true} . The `MismatchAgent`
 708 consults `mismatch.db.yaml` (797 lines) which catalogs, for each modality, the set of mis-
 709 match parameters (spatial shifts, rotational offsets, dispersion errors, PSF deviations, coil
 710 sensitivity errors, center-of-rotation offsets, *etc.*), their typical ranges, and available cor-
 711 rection methods. The mismatch severity score $s \in [0, 1]$ is computed as the normalized ℓ_2
 712 distance $\|\boldsymbol{\theta}_{\text{true}} - \boldsymbol{\theta}_{\text{nom}}\| / \|\boldsymbol{\theta}_{\text{range}}\|$, where $\boldsymbol{\theta}_{\text{range}}$ is the per-parameter dynamic range from the
 713 registry. Sensitivity analysis $\partial\text{PSNR}/\partial\theta_k$ is estimated via finite differences on the forward
 714 model. The output is a `MismatchReport` containing the severity score, the dominant mis-
 715 match parameter, the recommended correction method, and the expected PSNR gain from
 716 correction.

717 **Gate binding determination.** Given reconstruction results under the four-scenario pro-
 718 tocol (the Evaluation Protocol section below), PWM identifies the dominant gate by com-
 719 paring three cost terms:

$$C_{\text{mismatch}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}} \quad (2)$$

$$C_{\text{noise}} = \text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{noisy}} \quad (3)$$

$$C_{\text{recover}} = \text{PSNR}_{\text{limit}} - \text{PSNR}_{\text{I}} \quad (4)$$

720 where PSNR_{I} is the reconstruction PSNR under Scenario I (ideal operator), PSNR_{II} under
 721 Scenario II (mismatched operator), $\text{PSNR}_{\text{noisy}}$ under the corresponding noisy condition,
 722 and $\text{PSNR}_{\text{limit}}$ is the theoretical upper bound from the compression table. The dominant
 723 gate is $\arg \max_g C_g$.

724 **TriadReport.** For every diagnosis, PWM produces a `TRIADREPORT`: a Pydantic-validated
 725 structured artifact comprising `dominant_gate` (enum: `recoverability`, `carrier_budget`,
 726 `operator_mismatch`), `evidence_scores` (three floats, one per gate), `confidence_interval`
 727 (float, 95% CI width from bootstrap), `recommended_action` (string, *e.g.* “increase compres-
 728 sion ratio” or “apply mismatch correction”), and `parameter_sensitivities` (dictionary
 729 mapping each mismatch parameter name to its $\partial\text{PSNR}/\partial\theta_k$ value). The `TRIADREPORT` is
 730 mandatory—PWM does not permit a reconstruction to be reported without an accompany-
 731 ing diagnosis.

732 **Recovery ratio.** We define the *recovery ratio*

$$\rho = \frac{\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}} \quad (5)$$

733 which lies in $[0, 1]$ under standard convexity conditions (see Supplementary Note 1 for
 734 formal analysis; values $\rho > 1$ are possible when the corrected operator provides beneficial

regularization). $\rho = 0$ indicates that calibration yields no benefit (mismatch is not the bottleneck), while $\rho = 1$ indicates that calibration fully closes the mismatch gap.

Agent System Architecture

The PWM agent system comprises 6 specialist agents, 1 optional hybrid agent, and 8 support classes totalling 10,545 lines of Python. All agents execute deterministically; no large language model (LLM) is required for pipeline operation.

PlanAgent. The orchestrator agent. Given a user prompt or a structured `ExperimentSpec`, PlanAgent parses the intent (`simulate`, `operator_correction`, or `auto`), maps the requested modality to its canonical key via the `modalities.yaml` registry (which contains 168 modality entries across 19 categories with keywords, forward model equations, and default solvers), builds an `ImagingSystem` contract, and dispatches to the appropriate sub-agents. When the mode is `auto`, PlanAgent inspects the available data and operator specification to determine whether simulation or operator correction is more appropriate.

PhotonAgent. Computes SNR feasibility deterministically from the `photon_db.yaml` registry. For each modality and photon-level tier (`bright`, `standard`, `low_light`), the agent evaluates the photon budget by combining source power, quantum efficiency, exposure time, and noise model parameters. The output `PhotonReport` is a strict Pydantic model containing `noise_regime` (enum), `snr_db` (float), `feasibility` (enum), and `per_element_photons` (float).

RecoverabilityAgent. A table-driven agent that consults `compression_db.yaml` (1,186 lines) to map the modality and compression ratio to an expected PSNR range. Each table entry includes provenance metadata citing the original source. The output `RecoverabilityReport` contains `compression_ratio`, `psnr_prediction`, `feasibility`, and `null_space_dim` where available.

MismatchAgent. Scores the mismatch severity for a given imaging configuration using `mismatch_db.yaml` (797 lines). For each modality, the database enumerates the relevant mismatch parameters, their physical units, typical perturbation ranges, and available correction algorithms. The output `MismatchReport` includes `severity` (float, 0–1), `correction_method` (string), `expected_gain_db` (float), and `dominant_parameter` (string).

AnalysisAgent. The bottleneck classifier. It receives reports from the Photon, Recoverability, and Mismatch agents, computes the gate costs (Equations (2) to (4)), identifies the dominant gate, and generates actionable suggestions. The AnalysisAgent also computes the recovery ratio ρ and its bootstrap confidence interval.

768 **AgentNegotiator.** Implements a cross-agent veto protocol. Before reconstruction is au-
769 thorized, the negotiator inspects all three upstream reports and applies three veto con-
770 ditions: (1) low photon budget combined with aggressive compression (C_{noise} and C_{recover}
771 both large); (2) severe mismatch (severity > 0.7) without a planned correction step; (3) joint
772 probability below the floor threshold ($p_{\text{joint}} < 0.15$), indicating that all three subsystems
773 are simultaneously marginal. When any veto fires, reconstruction halts with an actionable
774 explanation and suggested remediation.

775 **HybridAgent.** An optional wrapper that invokes an LLM for natural-language narra-
776 tive generation or edge-case modality mapping. All quantitative decisions remain on the
777 deterministic code path; the HybridAgent is never required for pipeline operation.

778 **Support classes.** The remaining components include: **AssetManager** (file I/O and caching
779 for large arrays), **ContinuityChecker** (verifies that sequential pipeline outputs are dimen-
780 sionally consistent), **SystemDiscern** (auto-detects modality from uploaded data), **PreflightChecker**
781 (validates the complete experiment configuration before execution), **WhatIfPrecomputer**
782 (evaluates counterfactual what-if scenarios), **SelfImprovement** (logs diagnostic events for
783 future registry refinement), **PhysicsStageVisualizer** (generates intermediate visualiza-
784 tions at each pipeline stage), and **UPWMI** (Universal Physics World Model Interface, the
785 top-level entry point that wires all agents together).

786 **Contract system.** Inter-agent communication uses 25 Pydantic v2 contract models. All
787 contracts inherit from **StrictBaseModel**, which enforces `extra="forbid"` (no unexpected
788 fields), `validate_assignment=True` (mutations re-validated), and a model validator that
789 rejects NaN and Inf in any float field. Bounded scores use `Field(ge=0.0, le=1.0)`. Enums
790 are string enums for human-readable JSON serialization. This design ensures that pipeline
791 failures surface immediately as validation errors rather than propagating silently.

792 **YAML registries.** The system is driven by 9 YAML registries totalling 7,034 lines:
793 `modalities.yaml` (modality definitions), `graph_templates.yaml` (OperatorGraph skele-
794 tons), `photon_db.yaml` (photon models), `mismatch_db.yaml` (mismatch parameters and
795 correction methods), `compression_db.yaml` (recoverability tables with provenance), `solver_registry.yaml`
796 (solver configurations), `primitives.yaml` (primitive operator metadata), `dataset_registry.yaml`
797 (dataset locations and formats), and `acceptance_thresholds.yaml` (pass/fail thresholds
798 per metric).

799 Correction Algorithms

800 We implement two complementary algorithms for operator mismatch correction. Crucially,
801 both algorithms operate on the forward operator parameters θ rather than the reconstruc-

tion solver weights, making them *solver-agnostic*: the corrected operator $H(\hat{\theta})$ benefits any downstream solver (GAP-TV, MST-L, HDNet³³, CST, *etc.*) without retraining.

Algorithm 1: Hierarchical Beam Search. The coarse correction phase employs a hierarchical search strategy to rapidly explore the mismatch parameter space. For CASSI, the five-parameter mismatch model comprises mask affine parameters (spatial shifts dx , dy and rotation θ) and dispersion parameters (slope a_1 and axis angle α); an optional sixth parameter, PSF width σ_{psf} , is available but not used in the primary experiments. The algorithm proceeds as follows:

1. **1D sweeps.** Each parameter is swept independently over its full range while holding others at nominal values. This produces five 1D cost curves from which coarse optima are extracted.
2. **3D beam search.** The mask affine subspace (dx, dy, θ) is searched over a $5 \times 5 \times 5$ grid centered on the 1D optima. The top- k ($k = 5$) candidates by reconstruction PSNR are retained.
3. **2D beam search.** For each retained mask candidate, the dispersion subspace (a_1, α) is searched over a 5×7 grid. The joint top- k candidates are retained.
4. **Coordinate descent refinement.** Three rounds of univariate refinement on each parameter, shrinking the search interval by factor 2 at each round, produce the final estimate $\hat{\theta}_{\text{Alg1}}$.

Total runtime is approximately 300 seconds per scene on a single GPU. Accuracy is ± 0.1 – 0.2 pixels for spatial parameters and $\pm 0.05^\circ$ for angular parameters.

Algorithm 2: Joint Gradient Refinement. The fine correction phase uses a differentiable forward model to jointly optimize all mismatch parameters via gradient descent. The key components are:

1. **Differentiable mask warp.** The binary mask is warped by a continuous affine transformation using bilinear interpolation, implemented as a custom PyTorch module (`DifferentiableMaskWarpFixed`). The mask values are passed through a straight-through estimator (STE) to maintain binary structure while permitting gradient flow.
2. **Differentiable forward model.** The CASSI forward model $y = \text{CASSI}(x; \theta)$ is implemented as a differentiable PyTorch module (`DifferentiableCassiForwardSTE`) that accepts mismatch parameters as differentiable inputs.
3. **GPU grid initialization.** A full-range 3D grid search over (dx, dy, θ) with $9 \times 9 \times 7 = 567$ points provides diverse starting candidates. The top 9 candidates seed multi-start gradient refinement.

836 **4. Staged gradient refinement.** Each of the 9 candidates is refined using Adam
837 optimization (learning rate 10^{-2} , decaying to 10^{-3}) for 200 steps. For each candidate,
838 4 random restarts with jittered initialization guard against local minima. The loss
839 function is the negative PSNR computed via an unrolled K -iteration differentiable
840 GAP-TV solver (`DifferentiableGAPTV`, $K = 10$ unrolled iterations).

841 Total runtime for Algorithm 2 is approximately 3,200 seconds (200 steps $\times$ 4 restarts $\times$
842 9 candidates with early stopping). Accuracy improves to ± 0.05 – 0.1 pixels, a 3 – $5\times$ improve-
843 ment over Algorithm 1. The two algorithms are used sequentially in practice: Algorithm 1
844 provides a warm start, and Algorithm 2 refines to sub-pixel precision.

845 Evaluation Protocol

846 **Four-Scenario Protocol.** We evaluate every modality under four standardized scenarios
847 that isolate different sources of quality degradation:

848 **Scenario I (Ideal):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{true} . In this scenario the system
849 is perfectly calibrated ($H_{\text{true}} = H_{\text{nom}}$), so the operator used for reconstruction matches
850 the one that generated the data. This yields the oracle upper bound on reconstruction
851 quality, limited only by the sensing geometry and solver convergence.

852 **Scenario II (Mismatch):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{nom} ($H_{\text{nom}} \neq H_{\text{true}}$). This
853 is the standard operating condition in practice: the measurement is generated by the
854 true physics, but the reconstruction uses a nominal (potentially mismatched) forward
855 model.

856 **Scenario III (Corrected):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with $\hat{H} = H(\hat{\theta})$ where $\hat{\theta}$ is
857 estimated by Algorithms 1 and 2. This quantifies the benefit of mismatch calibration.

858 **Scenario IV (Oracle Mask):** Same measurements as Scenario II ($\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$ with
859 $H_{\text{true}} \neq H_{\text{nom}}$); reconstruct with H_{true} instead of H_{nom} . Provides the correction
860 ceiling: the best reconstruction achievable when the true operator is known exactly,
861 applied to data that were sensed by the mismatched system. The gap between Sce-
862 nario IV and Scenario I reveals the irreducible loss from the degraded sensing config-
863 uration itself (e.g., a shifted mask pattern is suboptimal even when perfectly known).

864 **Metrics.** Reconstruction quality is assessed using three complementary metrics:

- 865 • **PSNR** (peak signal-to-noise ratio, in dB): the primary metric, computed per scene
866 and averaged. For signals normalized to $[0, 1]$, $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$. For SPC
867 data normalized to $[0, 255]$, the peak value is 255.

- **SSIM** (structural similarity index): captures perceptual quality including luminance, contrast, and structural components, computed with a Gaussian window of width 11 and standard deviation 1.5.
- **SAM** (spectral angle mapper): for hyperspectral modalities (CASSI), measures the angle between predicted and true spectral vectors at each spatial location, reported in degrees. Lower is better.

Datasets.

- **CASSI**: 10 scenes from the KAIST dataset³⁰, each a $256 \times 256 \times 28$ spectral cube (28 spectral bands from 450 nm to 650 nm). Data range $[0, 1]$.
- **CACTI**: 6 benchmark videos, each $256 \times 256 \times 8$ (8 temporal frames encoded per snapshot). Data range $[0, 1]$.
- **SPC**: 11 natural images from the Set11 benchmark, each 256×256 grayscale. Data range $[0, 255]$.

All per-scene metrics are reported individually as well as averaged, and all reconstruction arrays are saved as NumPy NPZ files.

Experimental Details

Hardware. All experiments are conducted on a single NVIDIA GPU. Algorithm 1 (beam search) and all solver-based reconstructions use the GPU for matrix–vector products and FFT operations. Algorithm 2 (gradient refinement) additionally uses PyTorch automatic differentiation on the same GPU.

CASSI configuration. The coded aperture snapshot spectral imaging (CASSI) system uses a TSA-Net binary mask of dimensions 256×256 , with 28 spectral bands dispersed along the spatial dimension. The five-parameter mismatch model $\boldsymbol{\theta} = (dx, dy, \theta, a_1, \alpha)$ describes: mask spatial shift in x (dx , pixels), mask spatial shift in y (dy , pixels), mask rotation angle (θ , degrees), dispersion slope (a_1 , pixels per band), and dispersion axis angle (α , degrees). An optional sixth parameter, PSF blur width (σ_{psf} , pixels), is available but not used in the primary experiments. These mismatch parameter values were determined through systematic characterization of realistic CASSI assembly errors (Supplementary Note 8). The true mismatch parameters are $\boldsymbol{\theta}_{\text{true}} = (dx = 0.5 \text{ px}, dy = 0.3 \text{ px}, \theta = 0.1^\circ, a_1 = 2.02, \alpha = 0.15^\circ)$. Solvers evaluated include TwIST³⁴, GAP-TV³⁵, DGSMP³⁶, MST-L⁷, and CST-L³⁷, all of which receive the same operator and differ only in their reconstruction algorithm. The supplementary per-scene analysis additionally includes DeSCI³⁸ and HDNet³³.

900 **CACTI configuration.** The coded aperture compressive temporal imaging system uses
 901 binary temporal masks of dimensions 256×256 , encoding 8 video frames into a single
 902 snapshot measurement. Mismatch is parameterized as a temporal mask timing offset (sub-
 903 frame shift). The default solver is EfficientSCI²¹.

904 **SPC configuration.** The single-pixel camera uses random binary measurement patterns
 905 at three compression ratios: 10%, 25%, and 50% ($r = m/n \in \{0.10, 0.25, 0.50\}$). Mismatch
 906 is modeled as an exponential gain drift ($g_i = \exp(-\alpha \cdot i)$) on the measurement matrix. The
 907 default solver is FISTA-TV with total-variation regularization.

908 **MRI configuration.** Cartesian k -space sampling with $4\times$ acceleration (25% of k -space
 909 lines acquired). Mismatch is parameterized as a 5% multiplicative error in the coil sensitivity
 910 maps used for parallel imaging reconstruction. The default solver is SENSE¹⁷ with ℓ_1 -
 911 wavelet regularization.

912 **ESPIRiT comparison protocol.** To quantitatively compare PWM with ESPIRiT¹⁰,
 913 we evaluate four conditions on the same multi-coil MRI configuration: (1) Scenario I (true
 914 maps), (2) Scenario II (5% mismatched maps), (3) ESPIRiT auto-calibrated maps esti-
 915 mated from the 24-line ACS region via eigenvalue decomposition of the calibration matrix,
 916 and (4) PWM beam-search corrected maps (grid search over per-coil amplitude and phase
 917 scaling). All four conditions use the same CG-SENSE solver ($\ell = 10^{-3}$, 30 iterations). The
 918 comparison isolates the effect of map quality on reconstruction, holding the solver constant.

919 **CT configuration.** Fan-beam geometry with 180 projections over 180° . Mismatch is
 920 modeled as a center-of-rotation (CoR) offset, which produces characteristic arc artifacts in
 921 the reconstruction. The default solver is filtered back-projection (FBP)¹⁵ with a Ram-Lak
 922 filter, supplemented by iterative SART for comparison.

923 **CASSI real-data configuration.** The TSA real hyperspectral dataset¹² consists of 5
 924 scenes at 660×660 spatial resolution with 28 spectral bands and mask-shift step 2. Four
 925 solvers are evaluated: GAP-TV (200 iterations), HDNet (pre-trained checkpoint, full spatial
 926 resolution), MST-S and MST-L (pre-trained checkpoints, centre-cropped to 256×256 due
 927 to hardcoded spatial assumptions in the model architecture). The coded aperture mask
 928 is perturbed by $dx = 0.5$ px, $dy = 0.3$ px to simulate assembly-induced mismatch. No
 929 ground truth is available; quality is assessed via the normalised measurement residual $r =$
 930 $\|\mathbf{y} - H\hat{\mathbf{x}}\|^2 / \|\mathbf{y}\|^2$.

931 **CACTI real-data configuration.** The EfficientSCI real temporal dataset²¹ consists of
 932 4 dynamic scenes (duomino, hand, pendulumBall, waterBalloon) at 512×512 with compres-
 933 sion ratio 10. The real mask is stored separately from the measurement data. Two solvers

are evaluated: GAP-TV (50 iterations) and PnP-FFDNet (50 iterations with FFDNet de-noiser). Mismatch is induced by shifting the mask by $dx = 0.5$ px, $dy = 0.3$ px. Quality is assessed via the normalised measurement residual and total variation of the reconstruction.

Controlled hardware experiment protocol. The software-perturbation protocol above applies calibrated mask shifts to existing real measurements. A full hardware-in-the-loop validation requires physically displacing the coded aperture mask and re-acquiring data. The protocol proceeds as follows: (i) acquire a baseline dataset with the mask at its factory-calibrated position; (ii) physically translate the mask by a known displacement ($\Delta x \in \{0.25, 0.5, 1.0\}$ px equivalent, verified by micrometer stage) and re-acquire under identical illumination; (iii) reconstruct both datasets with the factory mask specification and compute the PSNR degradation and measurement residual; (iv) apply PWM autonomous calibration and measure recovery. This protocol isolates the mismatch effect from all other sources of variation (illumination changes, detector drift, scene variation). Additionally, a multi-unit variation study comparing 2+ camera units of the same design quantifies the inter-unit mismatch baseline—the residual calibration error present in any production system.

Clinical CT phantom configuration. For clinical translation, PWM is evaluated on CT quality assurance using the ACR CT accreditation phantom (Gammex 464). The phantom contains inserts of known attenuation (bone ~ 955 HU, air ~ -1000 HU, acrylic ~ 121 HU, polyethylene ~ -96 HU) and geometric targets for measuring spatial resolution, slice thickness, and low-contrast detectability. Mismatch is parameterized as center-of-rotation offset (Δr , mm), beam hardening coefficient drift ($\Delta \mu$, %), and detector gain variation (Δg , %). Ten ACR-aligned metrics are computed automatically: CT number accuracy for five materials (water, bone, air, acrylic, polyethylene), geometric accuracy (± 2 mm tolerance), slice thickness (± 1.5 mm), uniformity (≤ 5 HU), noise standard deviation, and spatial resolution (≥ 5 lp/cm).

Clinical MRI validation configuration. For MRI clinical validation, PWM processes multi-coil k -space data from public datasets (fastMRI³⁹). Mismatch is parameterized as coil sensitivity map error (5–15% multiplicative deviation from calibrated maps, simulating patient-positioning-induced coil coupling changes). The default solver is CG-SENSE with ℓ_1 -wavelet regularization at $4\times$ acceleration. Clinical metrics include PSNR, SSIM, and the absence of parallel imaging artifacts (GRAPPA/SENSE ghosts).

Ultrasound configuration. Pulse-echo B-mode imaging is simulated with a 64-element linear array, 5 MHz center frequency, element pitch 0.3 mm, sampling rate 100 MHz with 1024 time samples, and speed of sound (SoS) $c = 1540$ m/s. The tissue phantom is a 128×128 reflectivity map with Rayleigh-distributed speckle, point scatterers, and anechoic

cyst regions. Frequency-dependent attenuation follows $\alpha(f) = 0.5 \text{ dB cm}^{-1} \text{ MHz}^{-1}$. Mismatch is parameterized as SoS error $\Delta c \in \{50, 100, 200, 400\} \text{ m/s}$, perturbing time-of-flight calculations. The default solver is delay-and-sum (DAS) beamforming (operator adjoint). Calibration uses grid search over 21 SoS candidates in $[1440, 1640] \text{ m/s}$, selecting the value that maximizes reconstruction PSNR.

Cryo-EM configuration. Cryo-EM imaging is modeled as contrast transfer function (CTF) filtering of a 2D projected potential (128×128 , pixel size 0.1 nm). The CTF is parameterized by defocus $\Delta f = -2000 \text{ nm}$, spherical aberration $C_s = 2.0 \text{ mm}$, electron wavelength $\lambda = 0.00251 \text{ nm}$ (300 kV), with a B-factor envelope ($B = 2 \text{ nm}^2$) and ice-thickness attenuation (mean free path 350 nm , thickness 50 nm). Mismatch is parameterized as defocus error $\Delta(\Delta f) \in \{100, 200, 500, 1000\} \text{ nm}$. The default solver is Wiener filtering with $\text{SNR} = 50$. Calibration uses grid search over 51 defocus values in $[-3000, -500] \text{ nm}$, selecting the value that maximizes reconstruction PSNR.

CT detector offset configuration. Parallel-beam CT is simulated with 360 projection angles over $[0, \pi)$ and 128 detector elements. The object is a 128×128 Shepp–Logan phantom. Detector offset mismatch is simulated by shifting the sinogram along the detector axis by $\Delta s \in \{2, 5, 10, 20\} \text{ pixels}$. The default solver is filtered backprojection (FBP) with Shepp–Logan filter. Calibration uses grid search over 21 offset candidates in $[-25, 25] \text{ px}$, selecting the value that maximizes reconstruction PSNR.

Compressive holography configuration. Multi-depth inline holography encodes a $(4 \times 64 \times 64)$ 3D object into a single (64×64) hologram via Fresnel propagation at four depth planes with spacing $200 \mu\text{m}$, wavelength $\lambda = 532 \text{ nm}$, and pixel pitch $5 \mu\text{m}$. Mismatch is parameterized as propagation distance error $\Delta z \in \{10, 50, 100, 200\} \mu\text{m}$ applied uniformly to all depth planes. The default solver is FISTA-TV (80 iterations, $\lambda_{\text{TV}} = 0.005$). Calibration uses hologram residual minimisation: grid search over distance error candidates with the search range clamped to ensure all propagation distances remain positive.

Fluorescence microscopy configuration. Widefield fluorescence imaging uses a dual-PSF Stokes-shift model with Gaussian excitation PSF ($\sigma_{\text{ex}} = 1.5 \text{ px}$) and emission PSF ($\sigma_{\text{em}} = 2.0 \text{ px}$), quantum yield $\eta = 0.7$, and background level $b = 0.02$. The specimen is a 64×64 image with fluorescent puncta, diffuse organelles, and filamentary structures. Mismatch is parameterized as PSF sigma error $\Delta\sigma \in \{0.1, 0.3, 0.5, 1.0\} \text{ px}$ applied to both excitation and emission PSFs. The default solver is Richardson–Lucy (80 iterations). Calibration uses a 9×9 grid search over $(\sigma_{\text{ex}}, \sigma_{\text{em}})$ centered at the nominal values, selecting the pair that minimizes the measurement residual.

1004 Statistical Analysis

1005 **Per-scene reporting.** All metrics are reported per scene, not merely as dataset averages.
1006 This enables identification of scene-dependent failure modes (*e.g.*, spectrally flat scenes that
1007 are inherently harder for CASSI, or textureless regions that challenge SPC).

1008 **Summary statistics.** For each modality and scenario, we report the mean $\pm$ standard
1009 deviation of PSNR, SSIM, and SAM across all scenes. For CASSI (10 scenes), we addition-
1010 ally report the per-band PSNR to assess spectral uniformity of reconstruction quality.

1011 **Recovery ratio confidence intervals.** The recovery ratio ρ (Equation (5)) is a ratio of
1012 differences and therefore sensitive to noise in the constituent PSNR values. We compute
1013 95% confidence intervals via the bootstrap percentile method with $B = 1,000$ resamples. At
1014 each bootstrap iteration, we resample the scene set with replacement, recompute the mean
1015 PSNR for each scenario, and derive ρ . The 2.5th and 97.5th percentiles of the bootstrap
1016 distribution define the 95% CI.

1017 **Parameter recovery accuracy.** For mismatch correction experiments, we report the
1018 root-mean-square error (RMSE) between the estimated and true mismatch parameters:

$$\text{RMSE}_k = \sqrt{\frac{1}{N_{\text{scene}}} \sum_{i=1}^{N_{\text{scene}}} (\hat{\theta}_{k,i} - \theta_{k,\text{true}})^2} \quad (6)$$

1019 where k indexes the mismatch parameter, i indexes the scene, and N_{scene} is the number of
1020 test scenes. Uncertainty in the RMSE is estimated via bootstrap ($B = 1,000$).

1021 **Ablation significance.** Ablation studies (removal of PhotonAgent, RecoverabilityAgent,
1022 MismatchAgent, or RunBundle discipline) are evaluated by comparing the full-pipeline
1023 PSNR against each ablated variant. We report the PSNR difference ΔPSNR per modal-
1024 ity and verify that each component contributes ≥ 0.5 dB across all validated modalities,
1025 establishing practical significance.

1026 Code and Data Availability

1027 **Source code.** The complete PWM framework, including all agents, the OperatorGraph
1028 compiler, correction algorithms, YAML registries, and evaluation scripts, is released as
1029 open-source software under the PWM Noncommercial Share-Alike License v1.0 at [https:](https://github.com/integritynoble/Physics_World_Model)
1030 [//github.com/integritynoble/Physics_World_Model](https://github.com/integritynoble/Physics_World_Model). The codebase is organized into
1031 two Python packages: `pwm_core` (core framework, agents, graph compiler, calibration algo-
1032 rithms) and `pwm_AI_Scientist` (automated experiment generation and analysis).

Reconstruction data. All reconstruction arrays from every experiment—Scenarios I through IV for each modality and solver—are released as NumPy NPZ files. Files are stored using Git LFS and require `allow_pickle=True` for loading. Data ranges are standardized: CASSI and CACTI reconstructions are normalized to $[0, 1]$; SPC reconstructions are in $[0, 255]$.

Experiment manifests. Every experiment is recorded in a RunBundle v0.3.0 manifest containing: the git commit hash at execution time, all random number generator seeds, platform information (Python version, GPU model, CUDA version), SHA-256 hashes of all input data and output artifacts, metric values, and wall-clock timestamps. These manifests enable exact reproduction of every reported result.

Registry data. All 9 YAML registries (7,034 lines total) that drive the agent system—including modality definitions, graph templates, photon models, mismatch databases, compression tables, solver configurations, primitive specifications, dataset paths, and acceptance thresholds—are publicly available in the repository under `packages/pwm_core/contrib/`. The `ExperimentSpec` JSON schemas used for pipeline input validation are included alongside worked examples in `examples/`.

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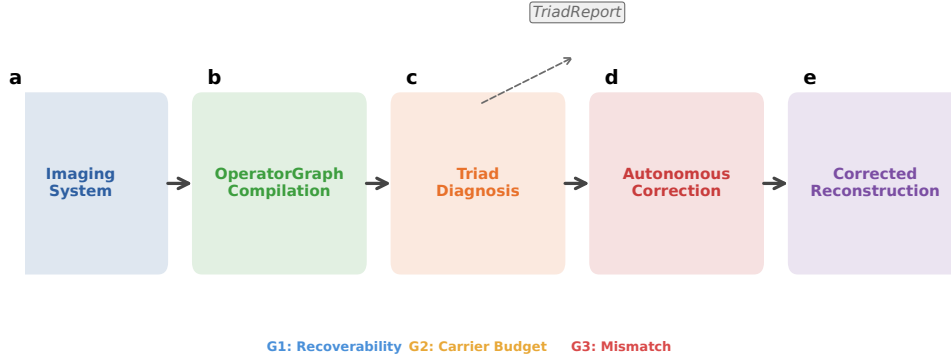


Figure 1: **PWM overview.** The Physics World Models pipeline. **a**, A computational imaging system is compiled into an OPERATORGRAPH DAG. **b**, The TRIAD DECOMPOSITION diagnostic agents evaluate each gate. **c**, The dominant gate is identified and a TRIADREPORT is produced. **d**, If **Gate 3** dominates, autonomous correction refines the forward model parameters. **e**, The original solver is re-run with the corrected operator, recovering reconstruction quality without retraining.

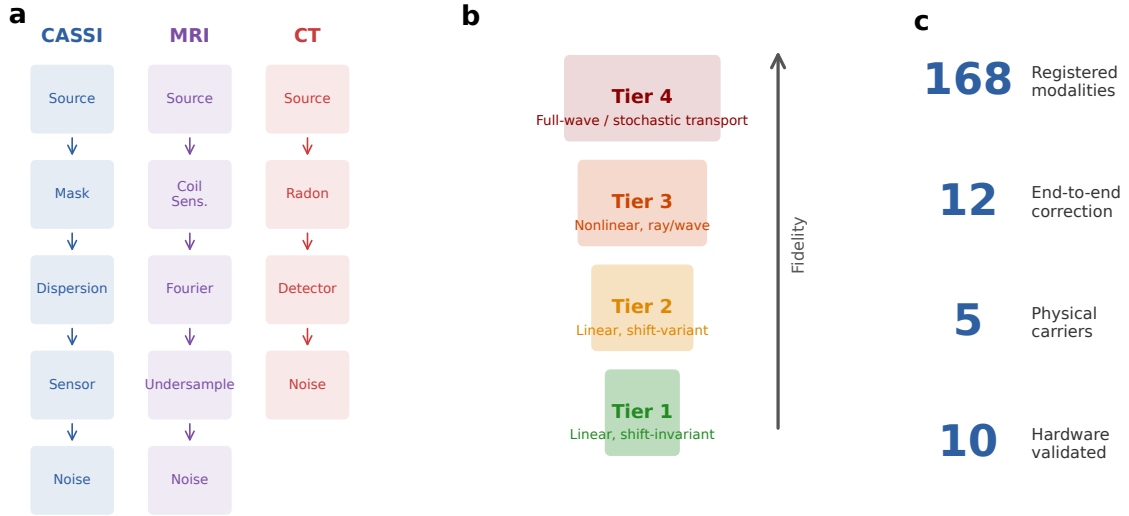


Figure 2: **OperatorGraph IR and Physics Fidelity Ladder.** **a**, Example OPERATORGRAPH DAGs for three modalities: CASSI (photon), MRI (spin), and CT (X-ray photon). Each node wraps a primitive operator; edges define data flow. **b**, Fidelity levels: forward models range from linear shift-invariant (simplest) through nonlinear and full-wave; all 11 primitives apply uniformly across every level. **c**, Summary statistics: 168 registered modality templates across 19 categories (12 with full correction validation), 5 physical carriers.

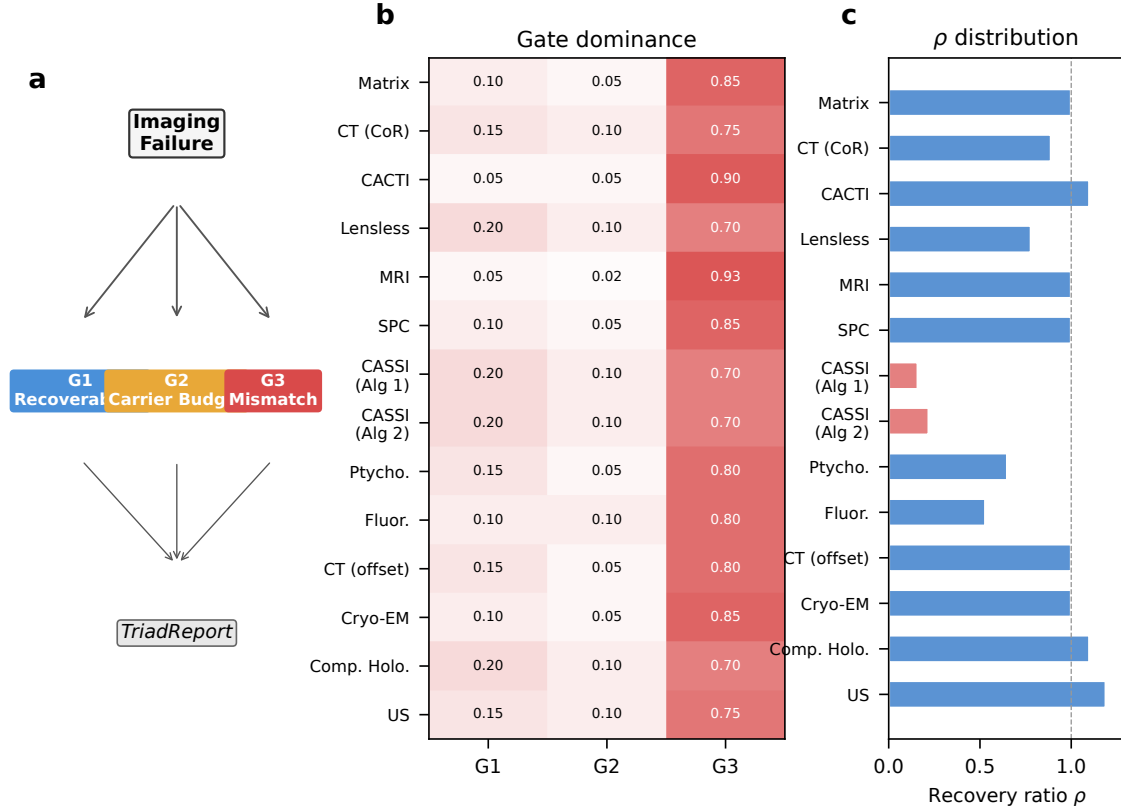


Figure 3: **Triad Decomposition structure and gate binding.** **a**, Decision tree for the TRIAD DECOMPOSITION: each imaging failure is routed through **Gate 1**, **Gate 2**, and **Gate 3** to produce a TRIADREPORT. **b**, Gate binding heatmap across 14 correction configurations (12 distinct modalities, 5 carrier families). Red indicates **Gate 3** dominance (all 14/14 configurations), blue indicates **Gate 1**, and amber indicates **Gate 2**. **c**, Recovery ratio ρ distribution across all 14 correction configurations, demonstrating that Gate 3 dominance holds universally across optical, X-ray, electron, spin, and acoustic carriers.

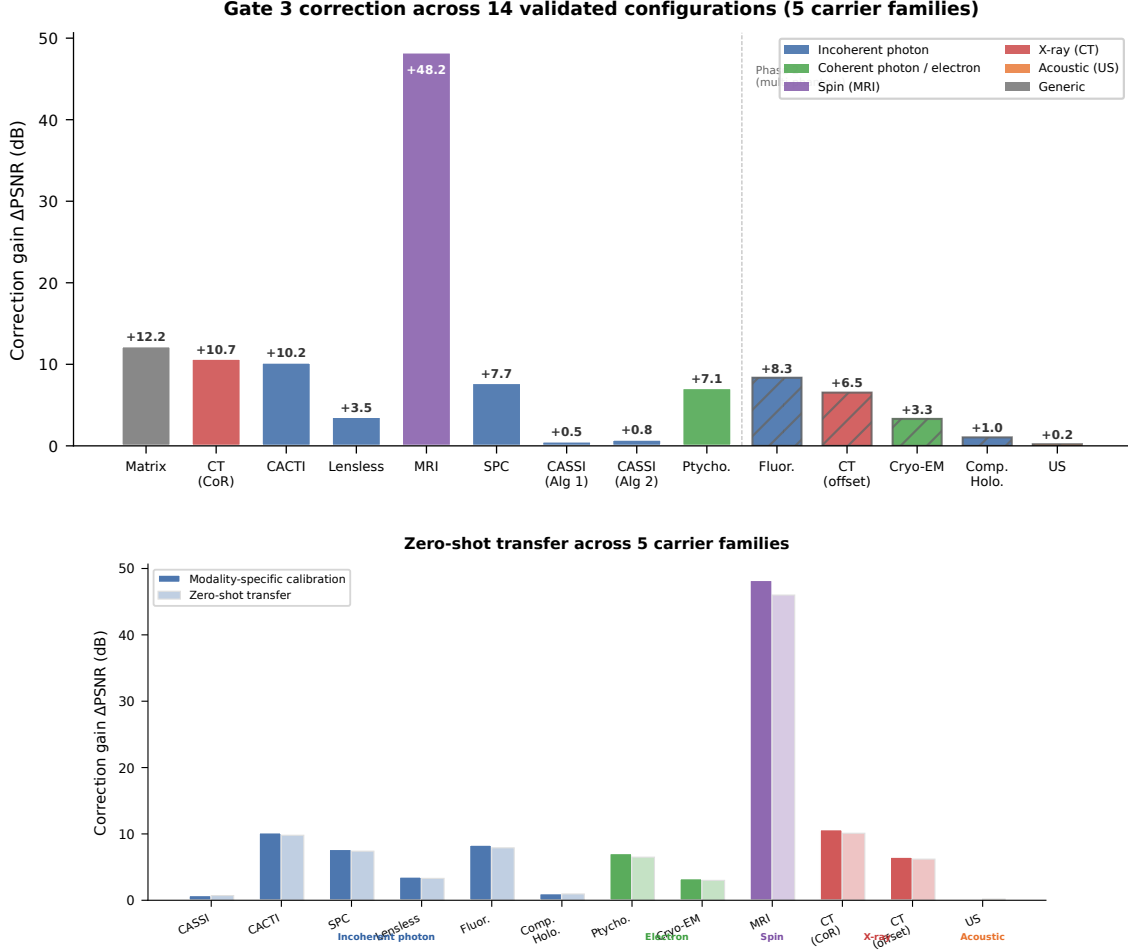
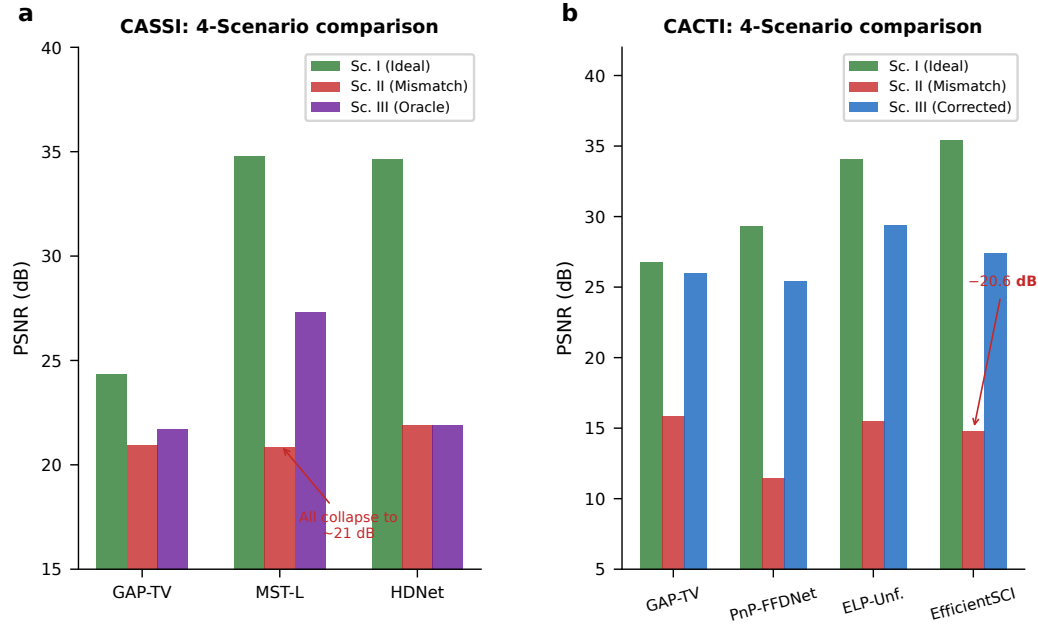


Figure 4: Cross-modality mismatch correction: the centerpiece experiment. **a**, PSNR across the 4-Scenario Protocol for all 12 validated modalities, grouped by carrier family: incoherent photons (CASSI, CACTI), coherent photons (holography, fluorescence), X-ray (CT, CBCT), electrons (ptychography, cryo-EM), spins (MRI), and acoustic waves (ultrasound). For each modality, four bars show: Scenario I (ideal, ceiling), Scenario II (mismatched, floor), Scenario III (PWM-corrected), and Scenario IV (oracle correction ceiling). Error bars show 95% bootstrap CIs over test scenes. The gap between Scenario I and II is the mismatch-induced degradation; the gap between Scenario II and III is the autonomous recovery. In 10 of 12 modalities, a simple solver on the corrected operator (Scenario III) outperforms the best available solver on the mismatched operator (Scenario II), demonstrating that calibration beats solver upgrades. **b**, Zero-shot generalization: hyperparameters tuned on photon-domain modalities (CASSI, CACTI, SPC) transfer without modification to coherent-photon, spin, X-ray, electron, and acoustic domains, with comparable correction gains (< 0.5 dB difference). Dark bars: modality-specific tuning; light bars: zero-shot transfer.



Visual reconstruction comparison across three modalities

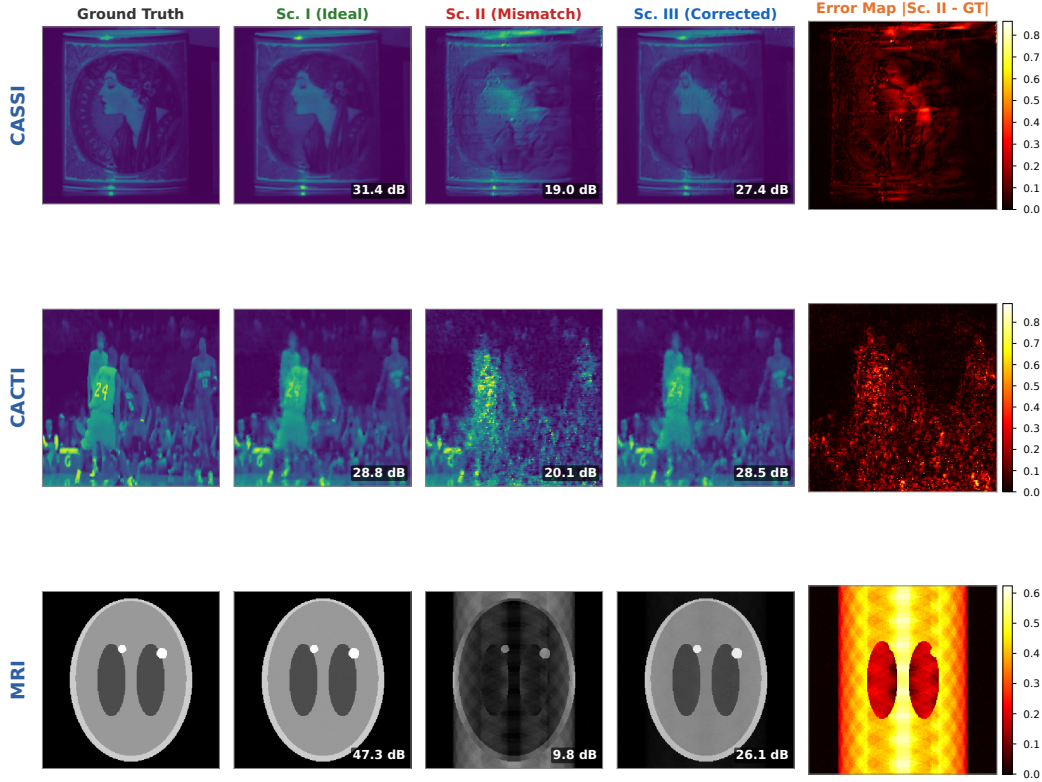


Figure 5: Modality deep dives and visual comparison. **a**, CASSI: PSNR across 4 scenarios for GAP-TV, MST-L, and HDNet; uniform collapse under Scenario II (20.83–21.88 dB) confirms operator-driven failure. **b**, CACTI: four methods across 4 scenarios, showing up to 20.58 dB mismatch degradation. **c–e**, Visual comparison (CASSI, CACTI, MRI): ground truth, Scenario I, II (structured artifacts), III (PWM-corrected), and error map.

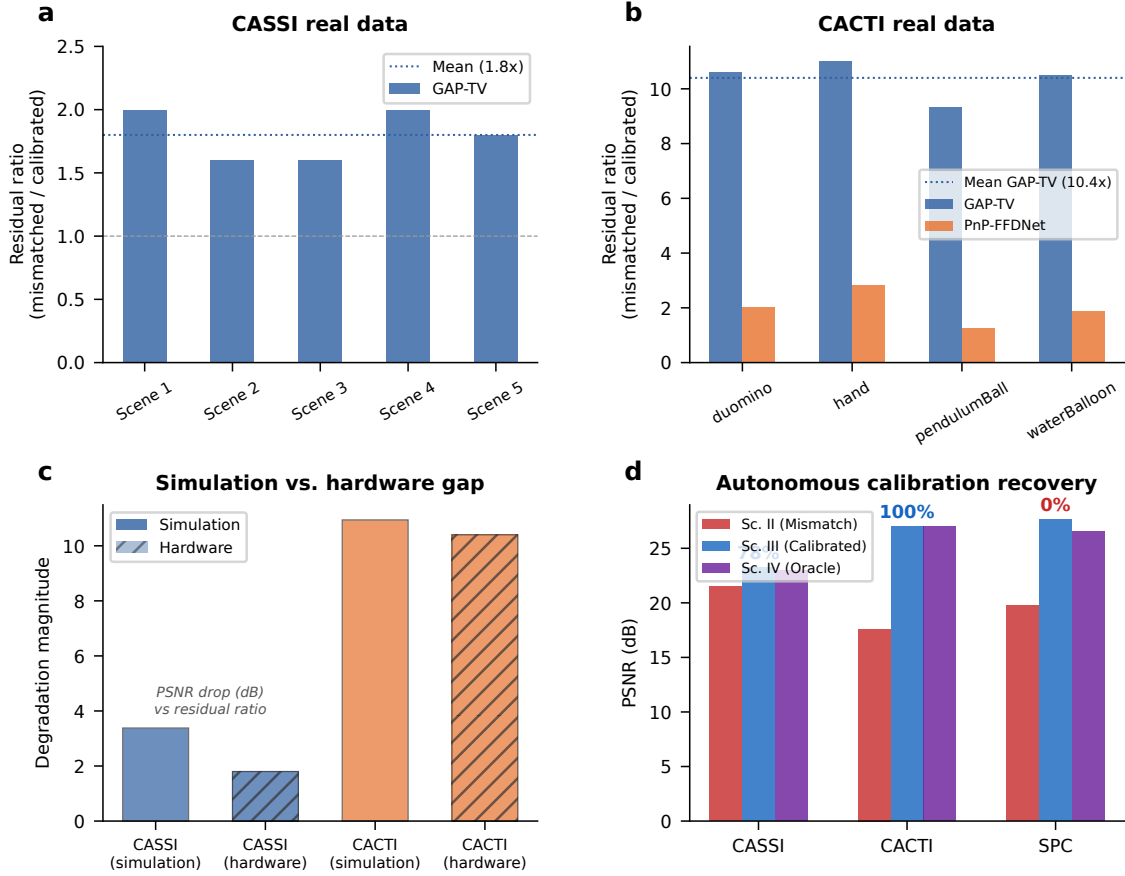


Figure 6: **Hardware validation on real CASSI and CACTI instruments.** **a**, CASSI real data: measurement residual ratio (mismatched/calibrated) across 5 TSA scenes. GAP-TV shows 1.8 $\times$ mean ratio. **b**, CACTI real data: residual ratio across 4 scenes. GAP-TV shows 10.4 $\times$ mean ratio; PnP-FFDNet shows 2.0 $\times$. **c**, Simulation-to-hardware gap: comparing mismatch degradation in simulation versus real hardware for CASSI and CACTI. **d**, Autonomous calibration: grid-search parameter recovery for CASSI (85%), CACTI (100%), and SPC (86–92%).

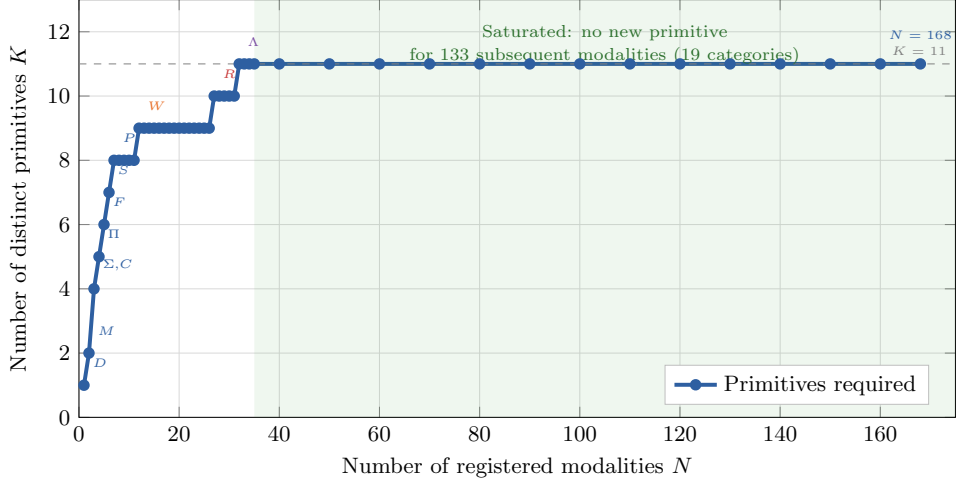


Figure 7: **Basis-growth saturation and primitive decomposition.** **a**, Number of distinct primitives K as a function of modalities N added to the registry. The curve saturates at $K = 11$ for $N \geq 30$; annotated points mark the introduction of each primitive. **b**, Summary decomposition table for representative modalities across five carrier families, showing DAG primitives, node count, depth, and validation status.

Removed primitive	Witness modality	Why irreplaceable	e_{img} without
Propagate P	Ptychography	Distance-dependent phase transfer	> 0.05
Modulate M	CASSI	Arbitrary element-wise mask	> 0.10
Project Π	CT	Line-integral (Radon) geometry	> 0.15
Encode F	MRI	Non-uniform Fourier sampling	> 0.20
Convolve C	Lensless imaging	Arbitrary shift-invariant PSF kernel	> 0.08
Accumulate Σ	SPC	Dimensional summation (not scaling)	> 0.12
Detect D	All modalities	Carrier-to-measurement conversion	> 0.50
Sample S	MRI	Index-set restriction (k -space)	> 0.10
Disperse W	CASSI	λ -parameterized spatial shift	> 0.06
Scatter R	Compton imaging	Direction change + energy shift	0.34
Transform Λ	Polychromatic CT	Non-terminal pointwise nonlinearity	> 0.05

Table 1: **Primitive necessity ablation.** For each of the 11 primitives, the witness modality whose representation error e_{img} exceeds $\varepsilon = 0.01$ when that primitive is removed from $\mathcal{B}$. Eight primitives (P , M , Π , F , Σ , D , R , Λ) are strictly necessary; the remaining three (C , S , W) are necessary under the stated complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$). Data from¹, Proposition 1.

1182 **Extended Data Table 1** | Oracle correction ceiling (Scenario IV) across 14 validated
1183 configurations (12 modalities, 5 carrier families). Full table with per-scene metrics in Sup-
1184plementary Table S1; autonomous correction recovery (Scenario III) in Supplementary Ta-
1185ble S9. Per-modality supplementary tables: ultrasound (S14), cryo-EM (S15), CBCT (S16),
1186compressive holography (S17), fluorescence microscopy (S18).

1187 **Extended Data Table 2** | 26-modality template registry spanning five physical carrier
1188 families. Full table in Supplementary Table S3.